



Bricks

A LEGO® BRICK MODEL OF

# VHS Cassette

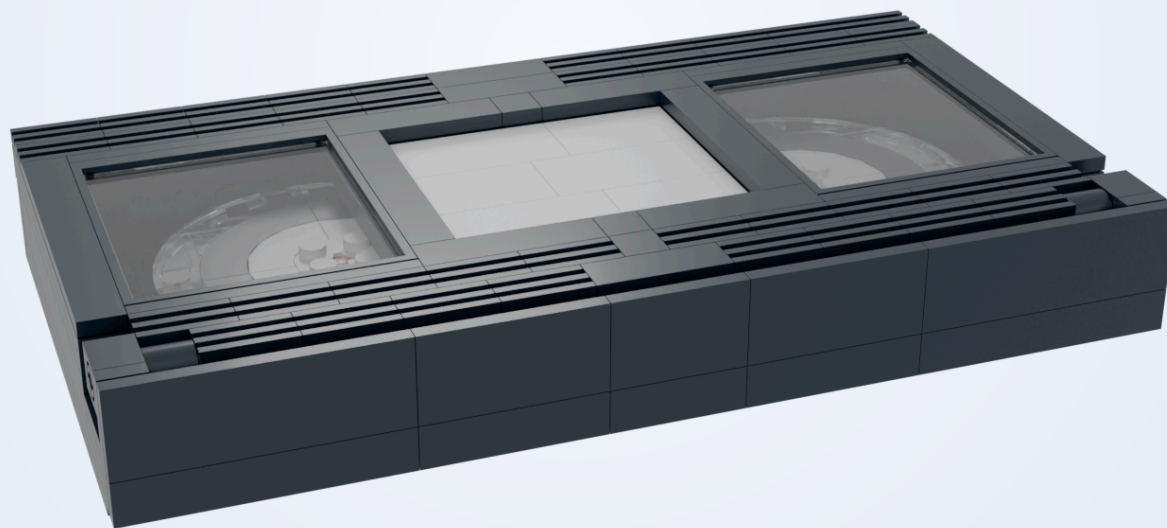
A 1:1 VHS tape, within a few millimetres of the real 187 x 103 x 25 mm cassette. Swing the dust flap up to see the tape; watch the reels turn through the windows.

**384** pieces · 139 steps · 14 sections

Unofficial fan-made building instructions

**Not physically build-tested.** Every part is a real LEGO element and the model was checked on a computer (connections, collisions, every step, balance, mechanism), but nobody has built it with real bricks yet.

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# Before you start

## An unofficial fan-made manual

This booklet is not a LEGO product. It was generated from the model file by a computer program and checked part by part.

## Not physically build-tested

The model and these steps were checked on a computer only: every part/colour is a real LEGO element, all 996 connections between parts were matched in 3D, no two parts overlap, every part can be pushed into place in its step, and the model's balance was estimated; the mechanism was swept through its whole range. Not tested: how firmly parts hold, part tolerances and how it copes with handling. Take your time and support the model while you work.

## The finished model

About 18.4 × 10.2 cm and 2.6 cm tall, about 313 g (estimated).

Built like a real cassette, from the inside out. First the chassis: the bottom with its drive holes and screw heads, the lamp tube, the reel-lock release, the guide rollers and the tape stretched across the mouth. Two reels drop onto their red spindles: the supply reel and the take-up reel. The end walls go on, then the top shell, built upside down with its grip bands, recessed label, spine label and two big windows, is lowered over everything. Last, the full-length dust flap hangs on two Technic pins.

## You will need

- The 384 pieces in 5 colours listed at the back, with their LEGO element IDs and BrickLink numbers.
- A brick separator for pieces that sit too deep for a finger.

## Tips

- Sort the pieces by colour before you start.
- Build the sections in order: each sub-assembly is finished before it is used.
- New pieces in a step have a yellow outline; earlier pieces are drawn paler.
- Each reel turns on its red Technic axle in the chassis' round holes. The axle's end shows in the drive hole underneath: turn it with a fingertip to wind the tape.
- Lower the top shell straight down so the window hangers find their ledges and the reel hubs slide under the windows.
- The write-protect tab on the back is a single tile: pull it off to protect your recording.

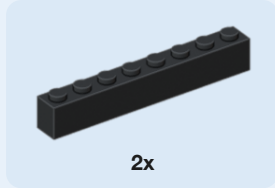
## The 5 colours

- |                                                                                                       |                                                                                                      |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|  Black: 335      |  Flat Silver: 5 |
|  White: 32       |  Red: 2         |
|  Trans-Clear: 10 |                                                                                                      |

# How to read the steps

## Parts and step number

7

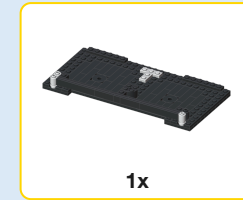


The parts for the step and how many of each, then the step number. The parts of a step are drawn at one scale, so a longer one looks longer; a tile or a long plate also has its size in studs under it.

## New parts

The parts added in a step have a yellow outline and are drawn in full colour; the parts from earlier steps are drawn paler.

## Sub-assemblies



A yellow-framed picture means: put in the sub-assembly you built in its own section. "Build 4x" means you need four of them.

## Sections

Each section starts with its name at the top of the page. Sections are built in the order they appear.

## Seen from below

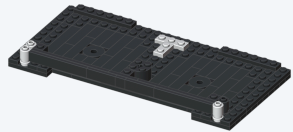
When a step adds parts underneath, the picture shows the model from below. A small label says so (and "seen from behind" likewise). A piece the picture cannot show gets a small picture beside it, ringed in yellow, from the side it can be seen from: where it clicks in.

## Insert direction

Some parts slide in sideways (axles through bearings, gears onto axles). The picture shows where they end up; push them in along their axle.

# Build overview

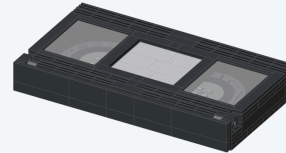
The model is built in 14 sections. Each is finished before it is used.



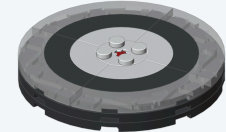
**1. Chassis**  
steps 1–53



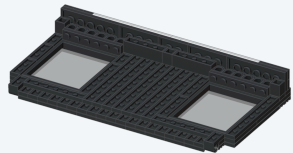
**2. Supply reel (full)**  
steps 54–58



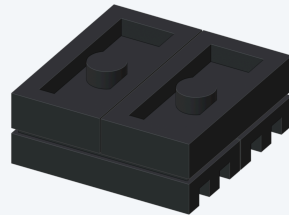
**3. VHS Cassette**  
steps 59, 65–66, 129, 139



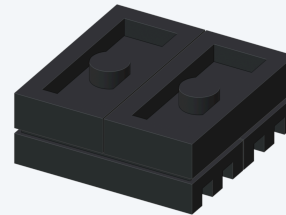
**4. Take-up reel**  
steps 60–64



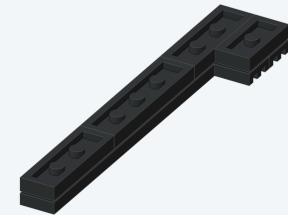
**5. Top shell**  
steps 67–87, 89, 91, 94–95, 97, 101–105, 111–122, 127–128



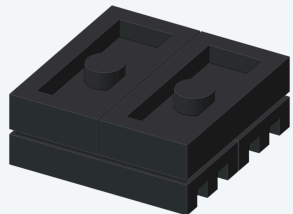
**6. Top shell: strip 1**  
step 88



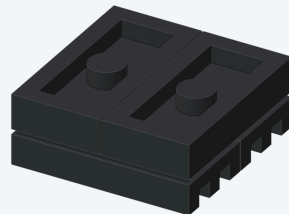
**7. Top shell: strip 2**  
step 90



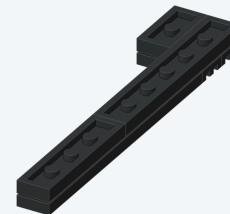
**8. Top shell: strip 3**  
steps 92–93



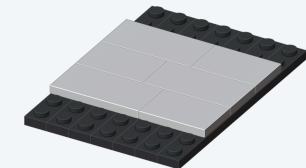
**9. Top shell: strip 4**  
step 96



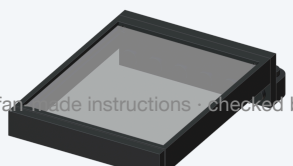
**10. Top shell: strip 5**  
step 98



**11. Top shell: strip 6**  
steps 99–100

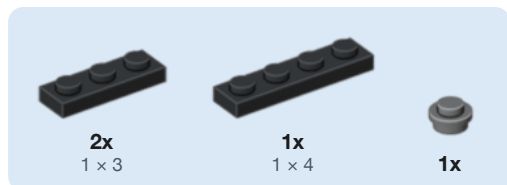


**12. Face label**  
steps 106–110

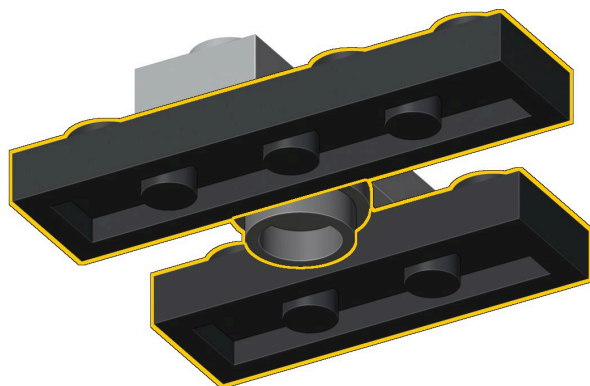
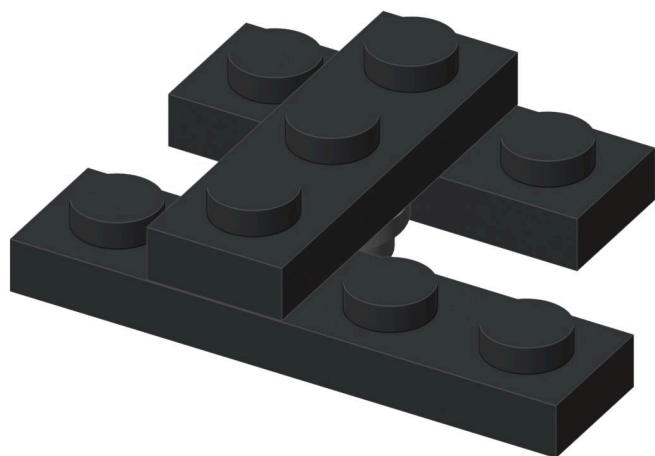


# CHASSIS

## 1

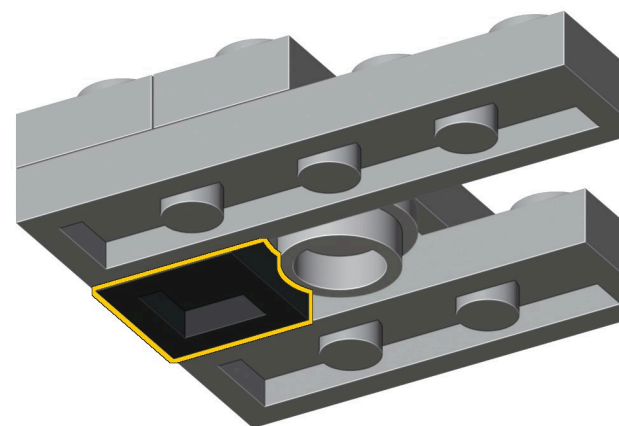
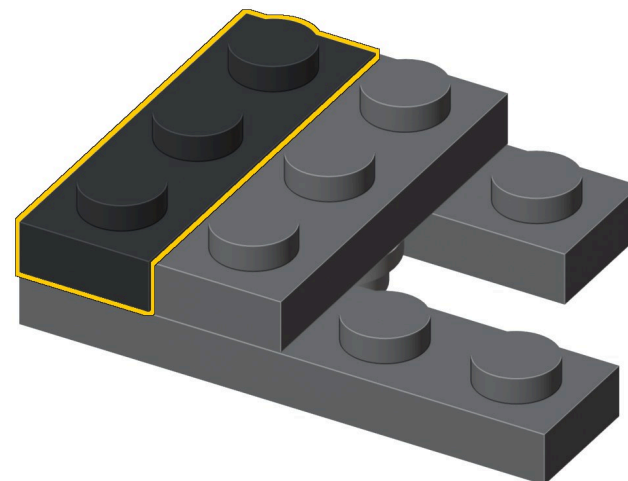
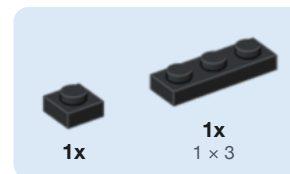


Chassis, bottom layer: the Technic plates' middle holes are the reel drive holes and the lamp hole; silver screw heads sit flush underneath



From below: the outlined pieces click in here.

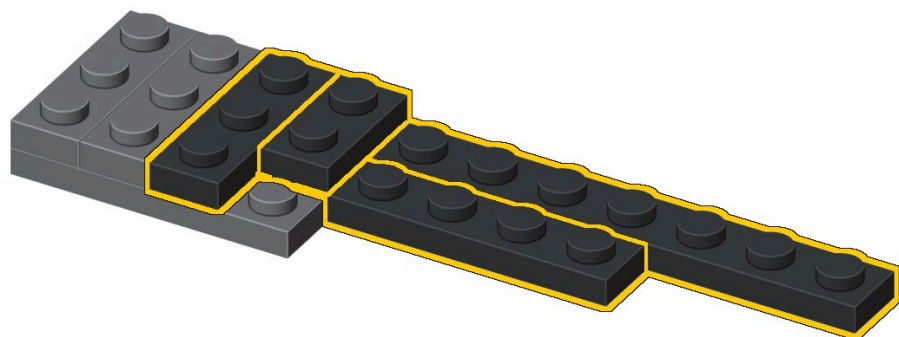
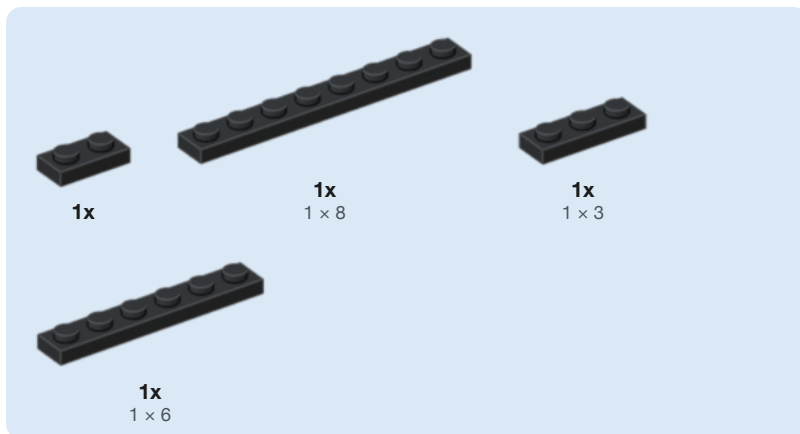
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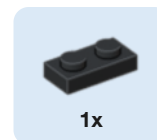
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Chassis

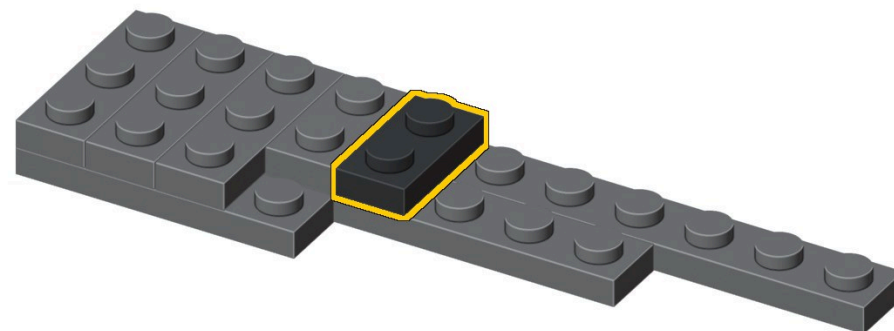
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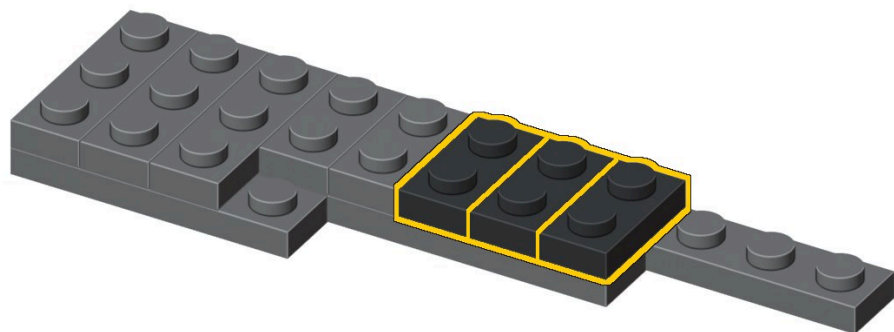
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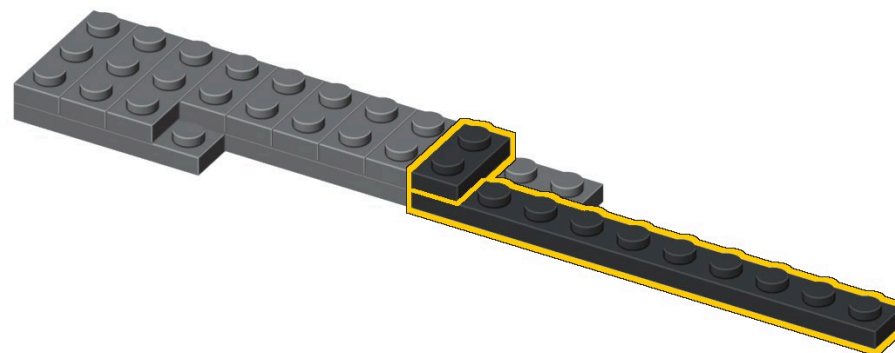
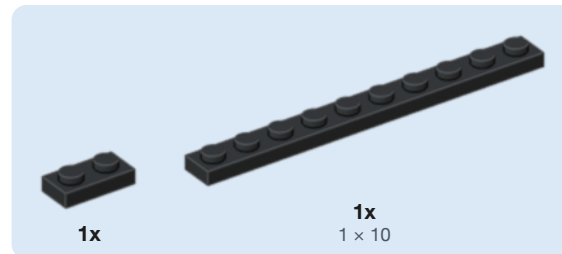
Chassis



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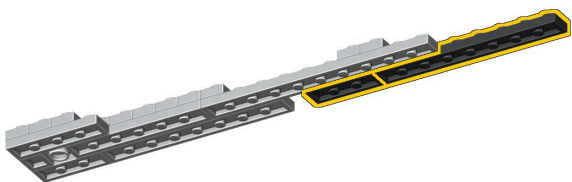
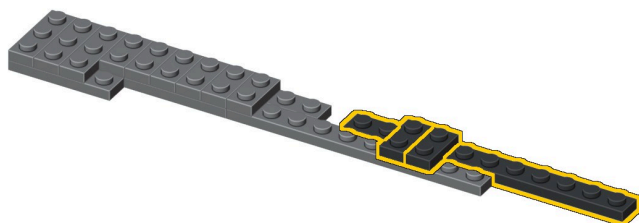
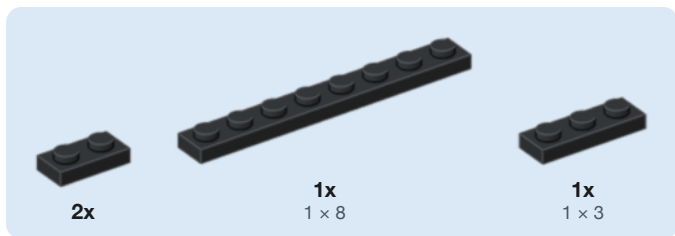


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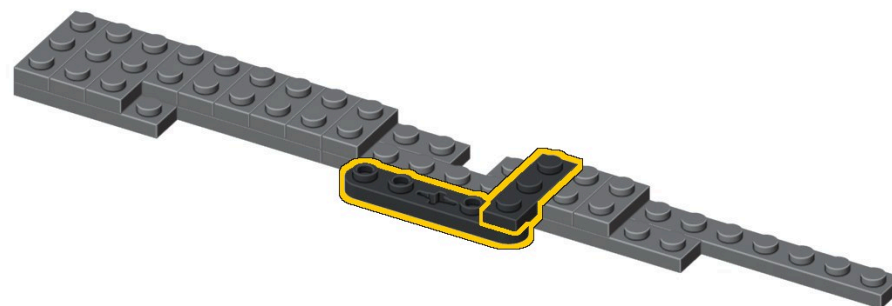
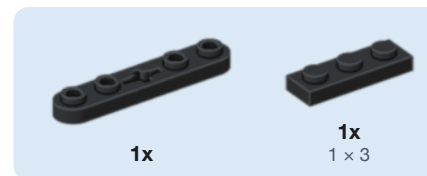
Chassis

# 7



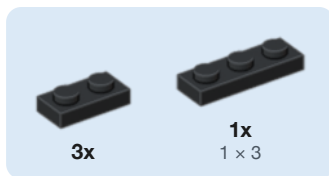
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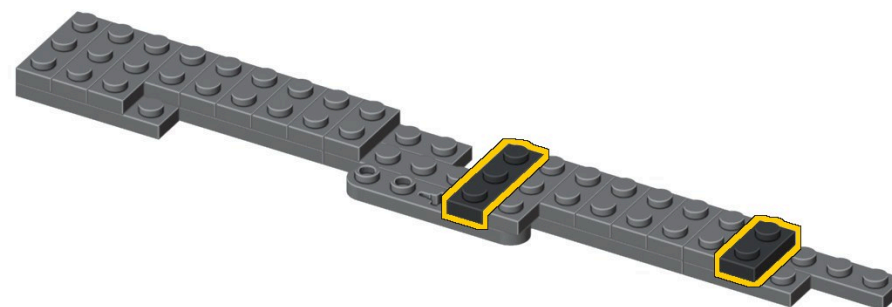
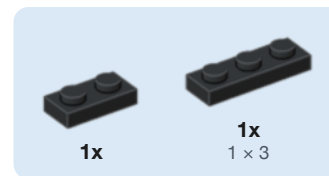


Chassis

9

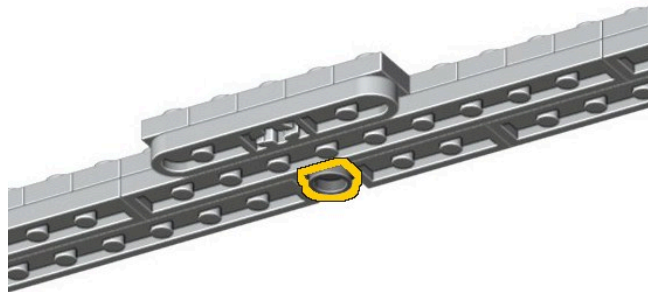
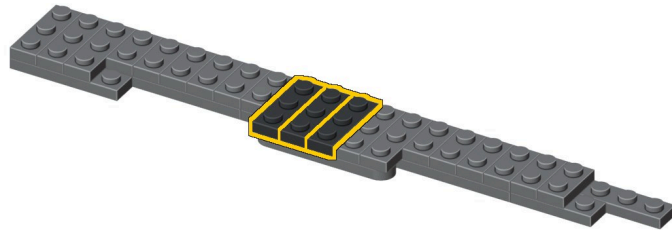
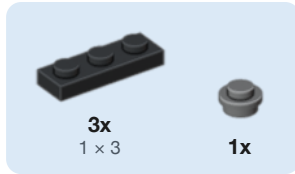


10



Chassis

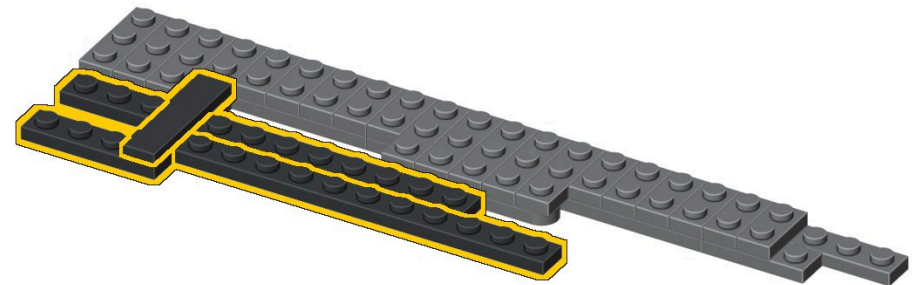
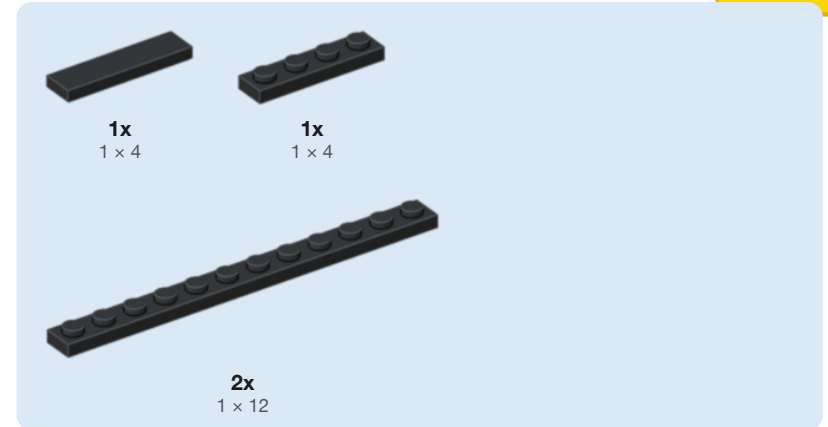
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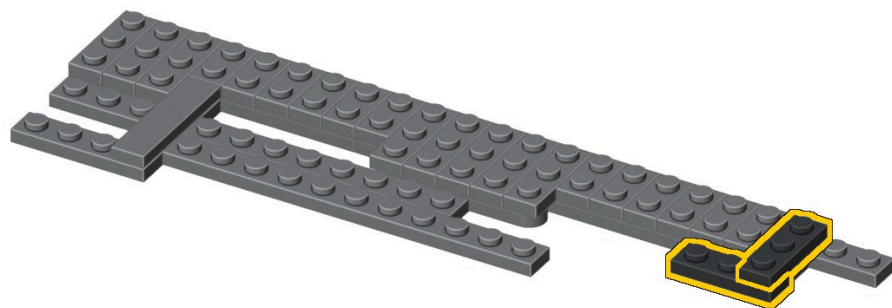
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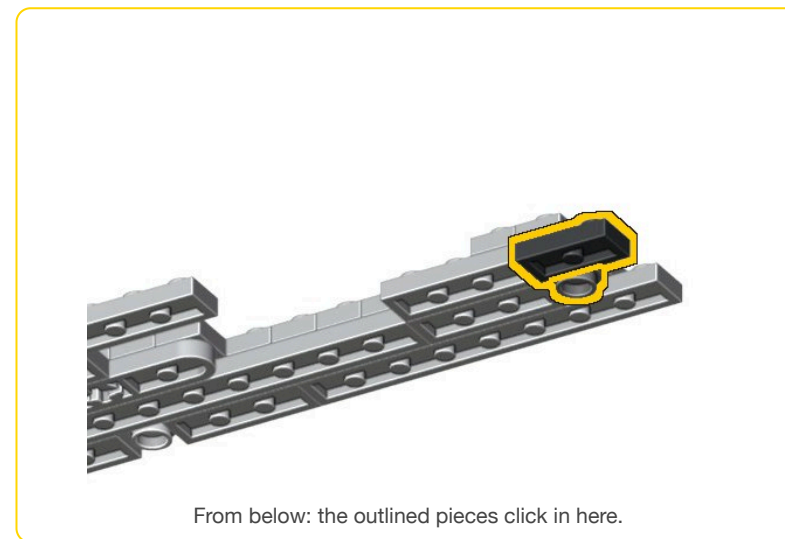
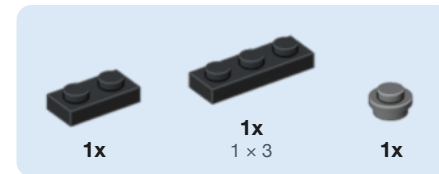
Chassis



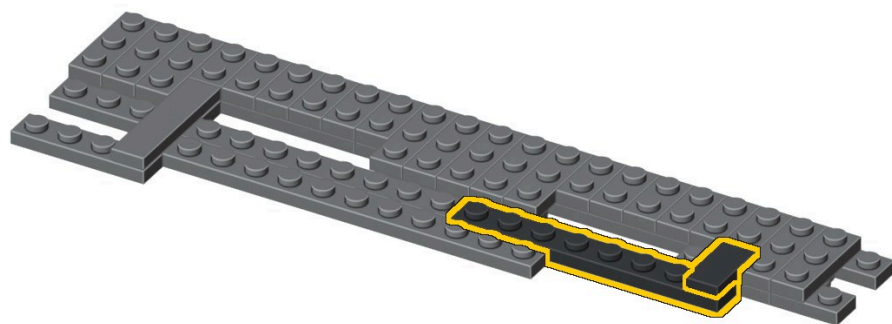
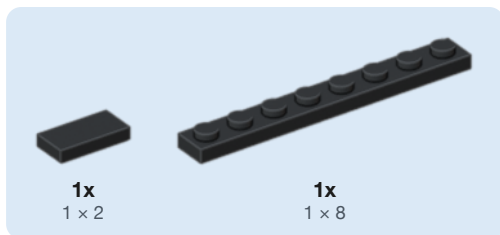
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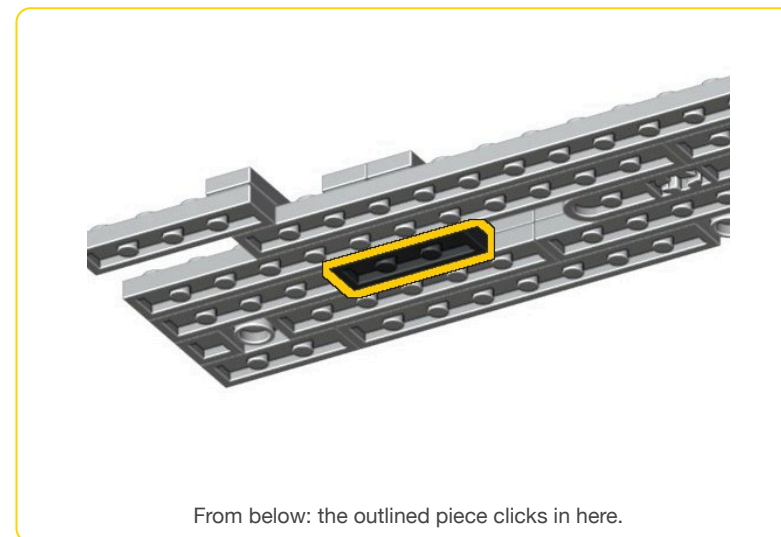
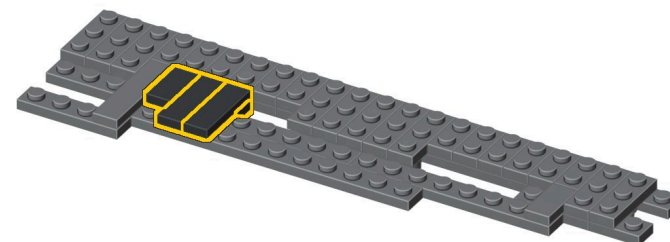
# 14



# 15

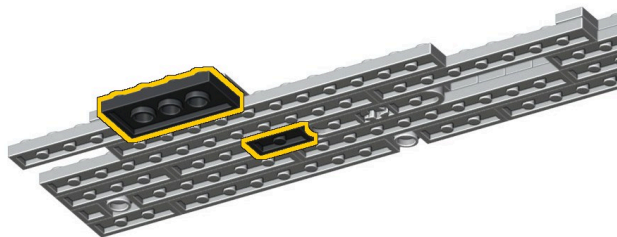
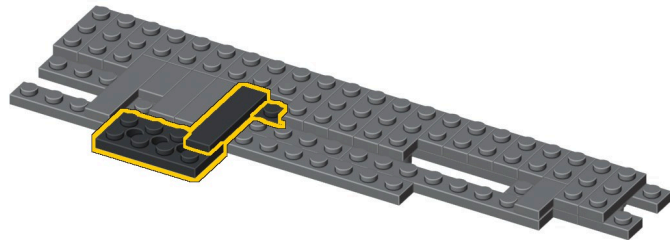
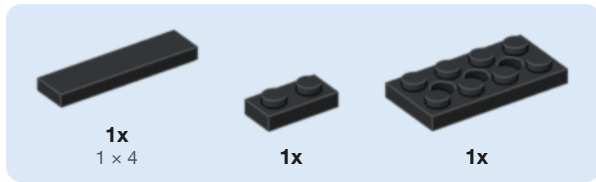


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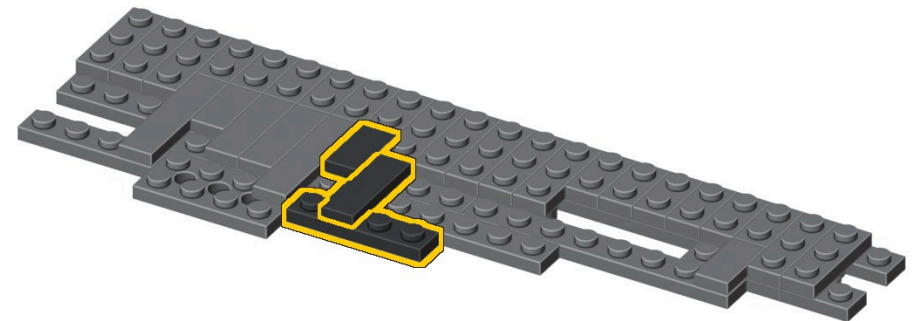
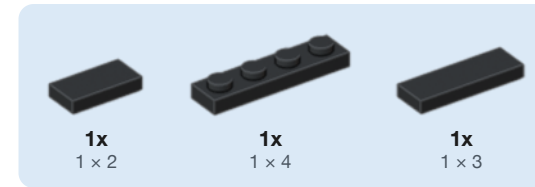
Chassis

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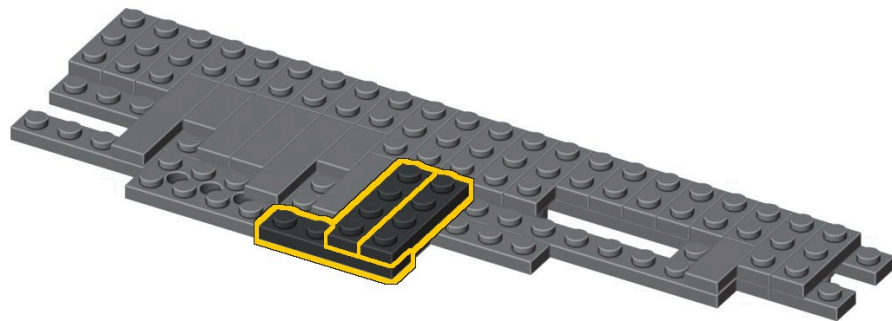
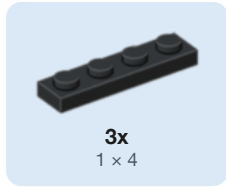
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# 18



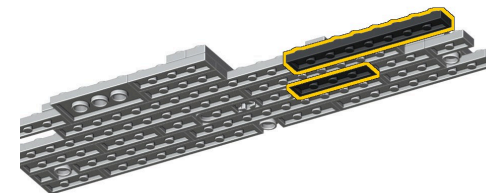
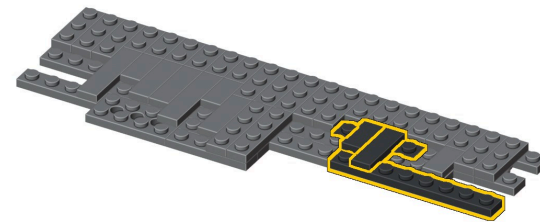
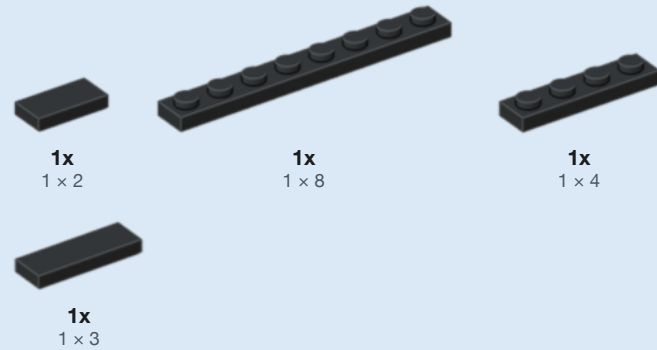
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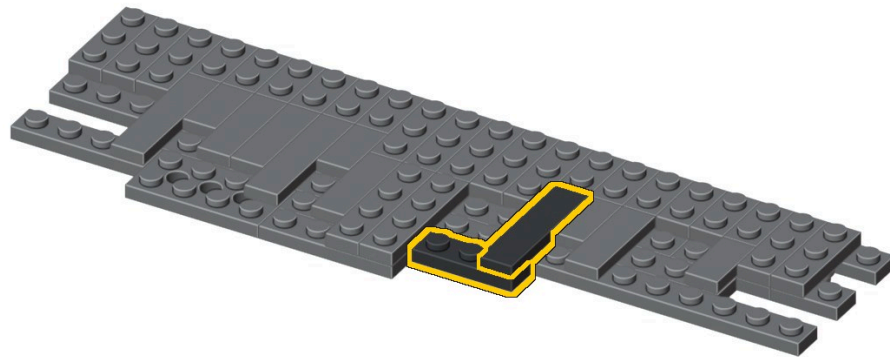
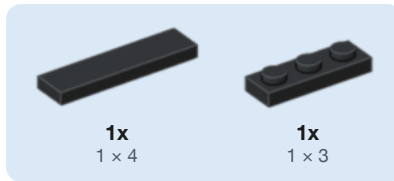
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Chassis



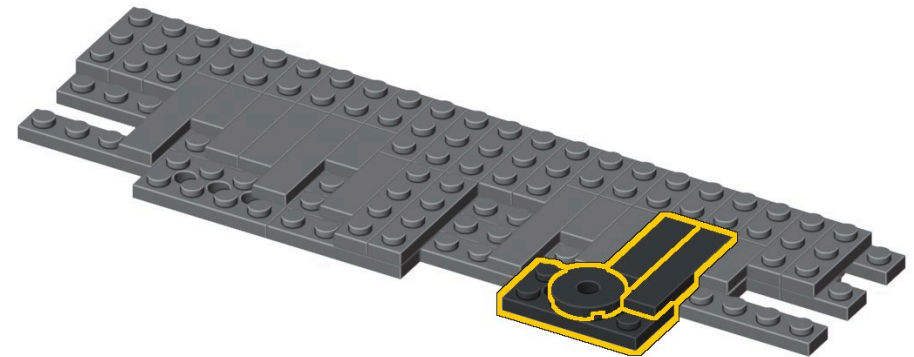
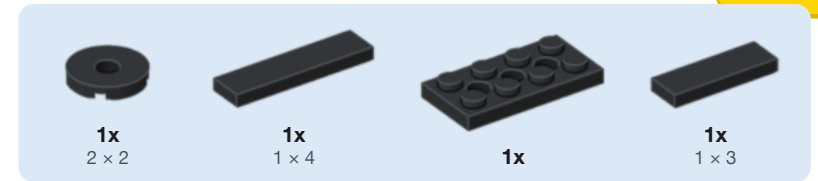
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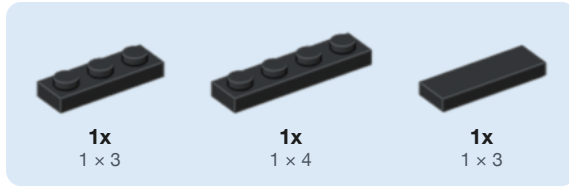


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Chassis

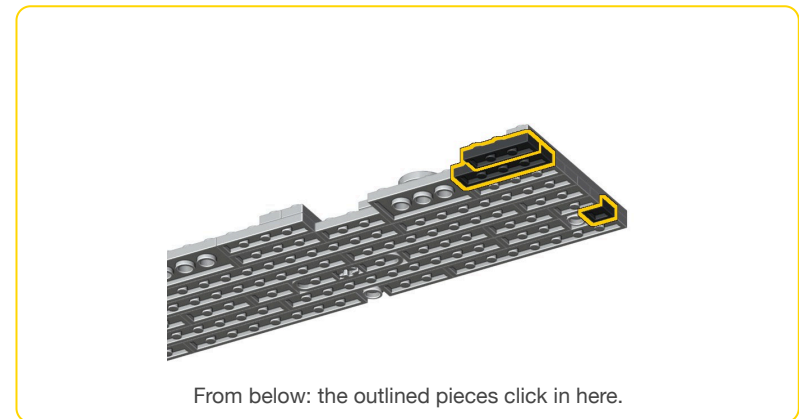
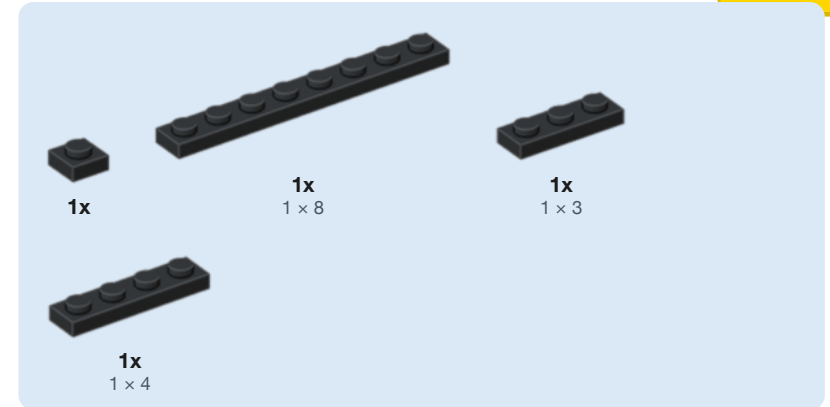


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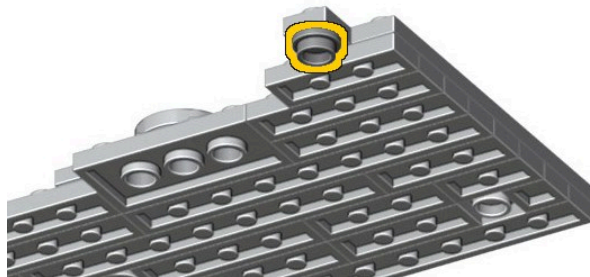
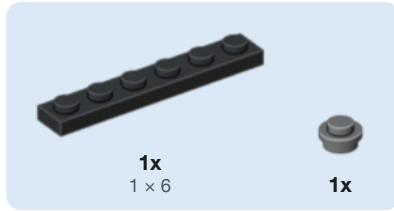


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Chassis

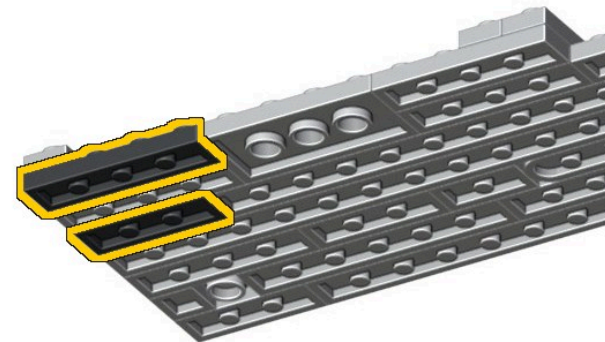
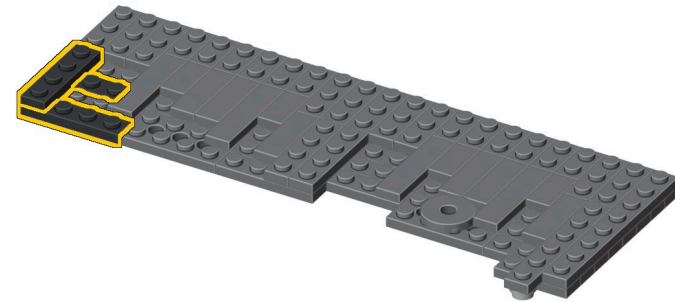
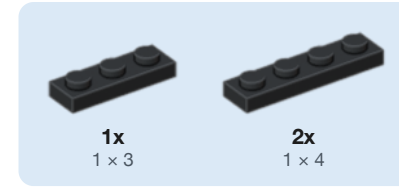


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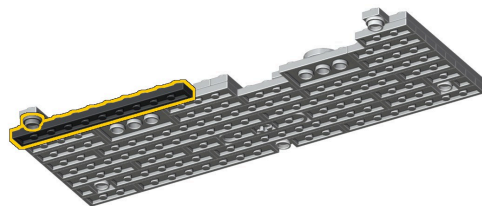
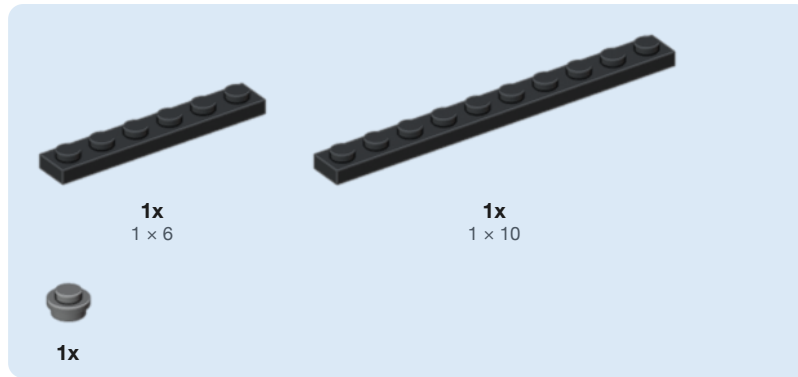
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# 26



From below: the outlined pieces click in here.

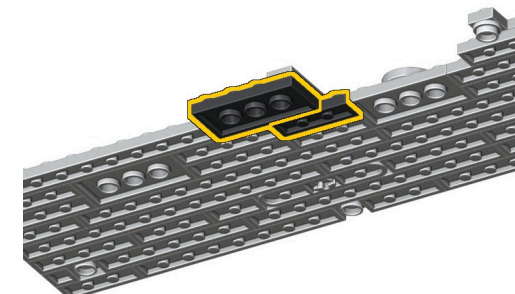
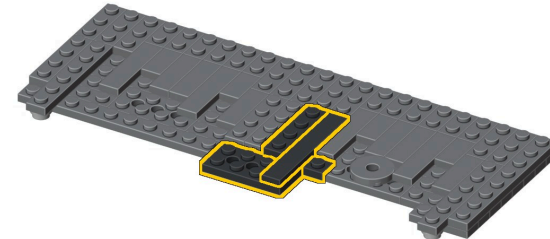
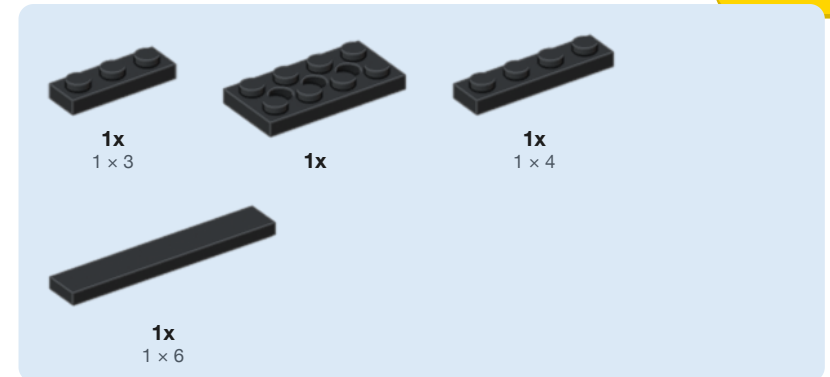
# 27



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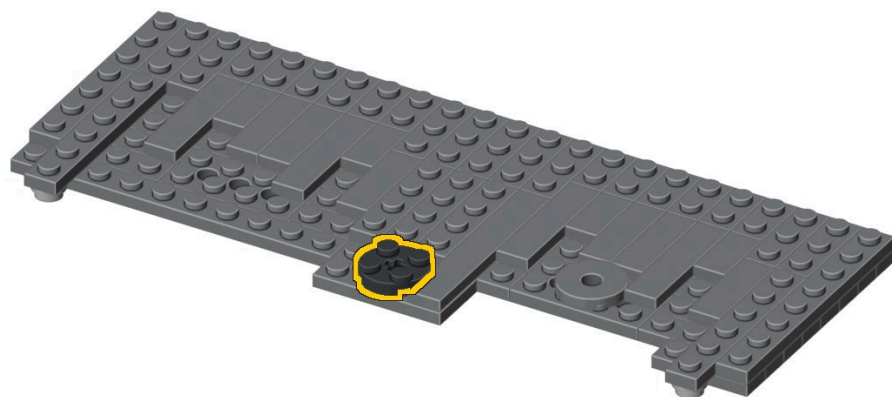
# 28

Chassis



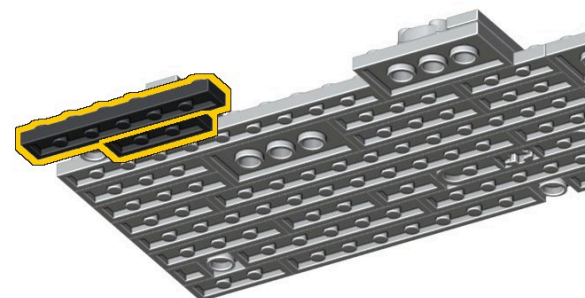
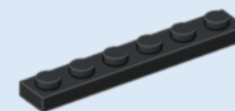
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# 29



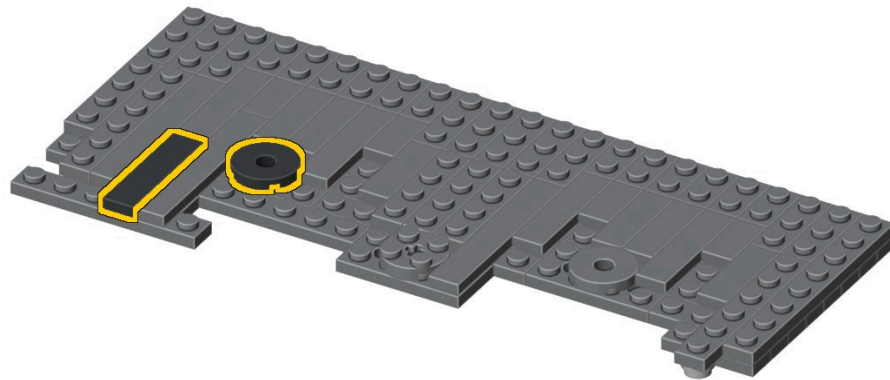
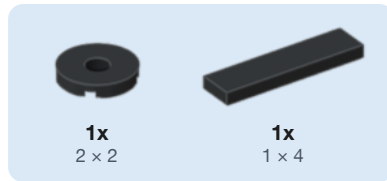
# 30

Chassis



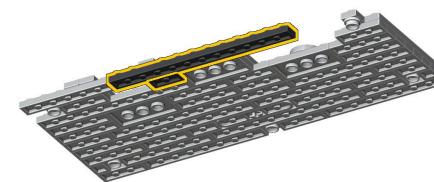
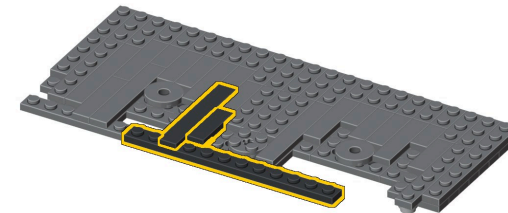
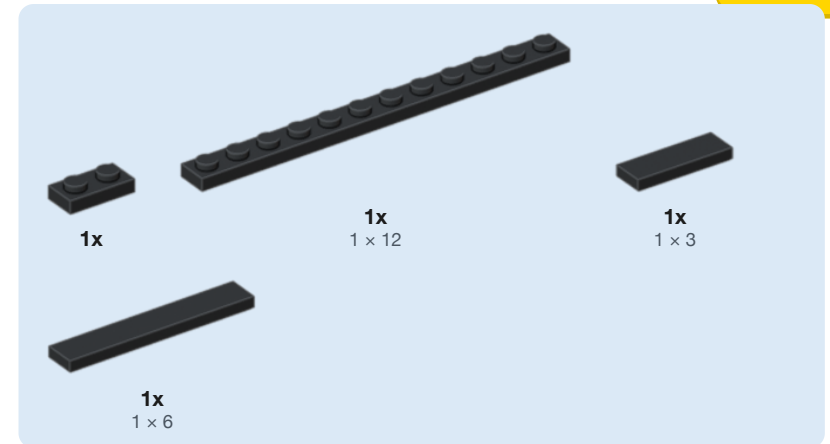
From below: the outlined pieces click in here.

# 31



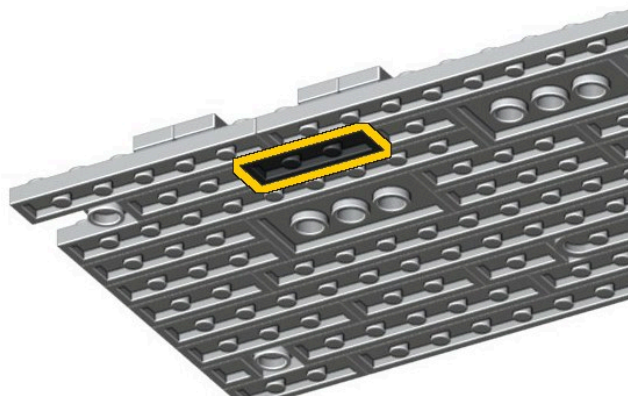
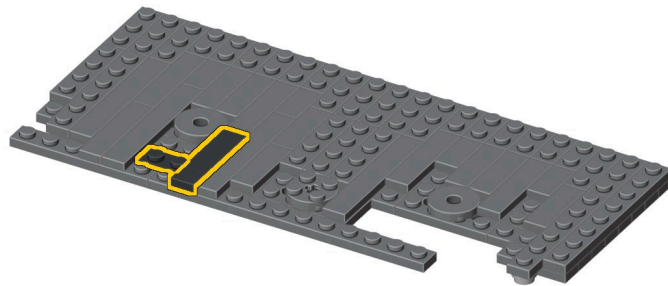
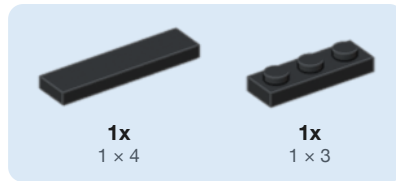
# 32

Chassis



From below: the outlined pieces click in here.

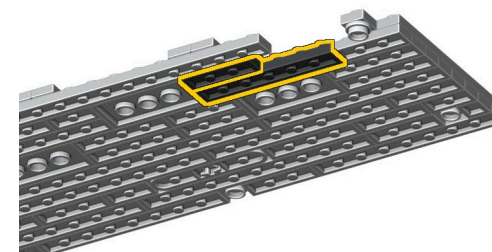
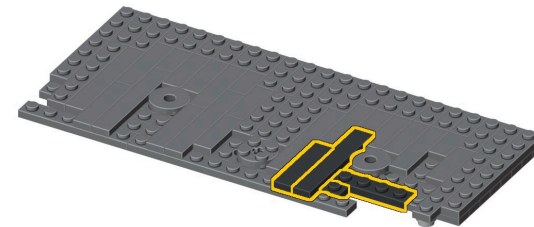
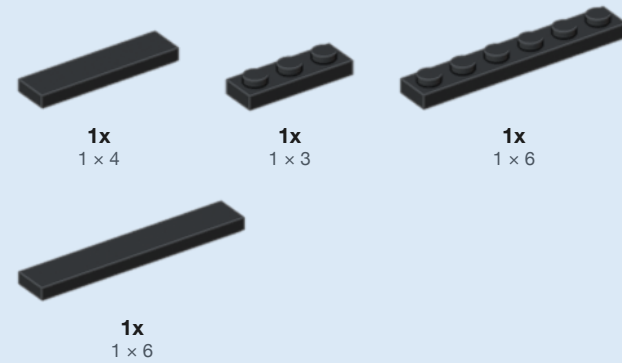
# 33



From below: the outlined piece clicks in here.

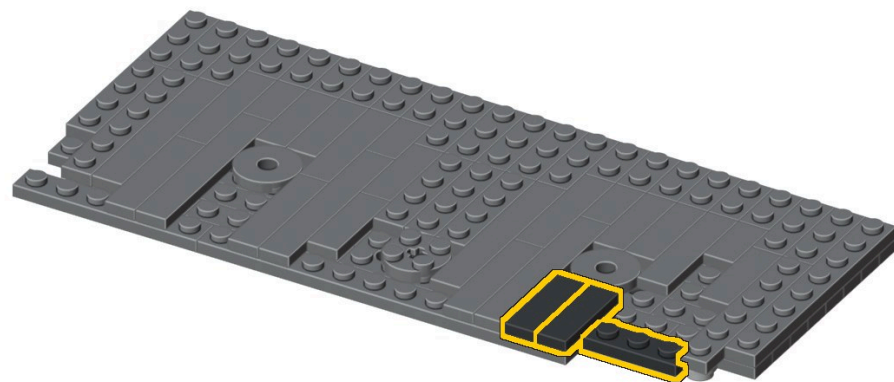
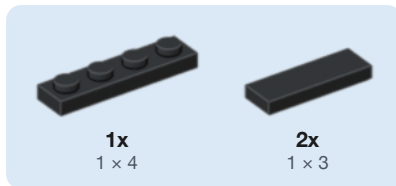
# 34

Chassis



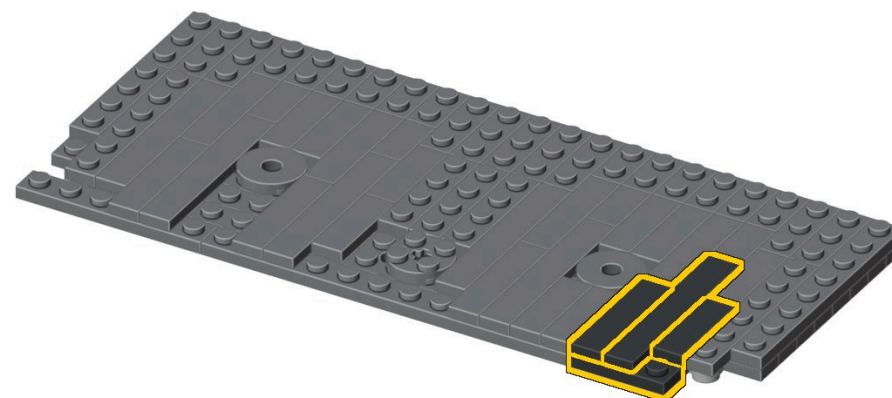
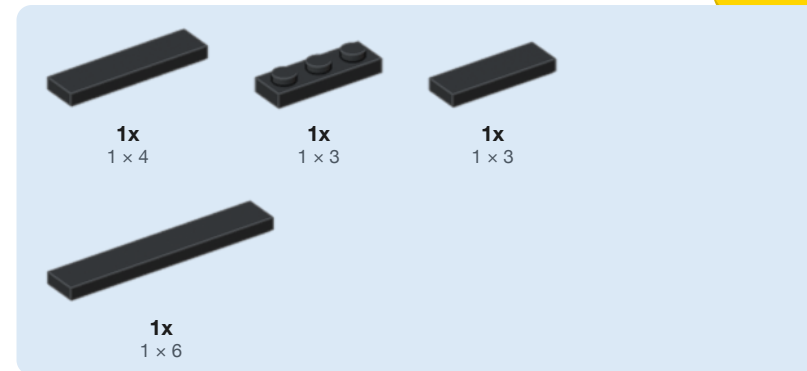
From below: the outlined pieces click in here.

# 35

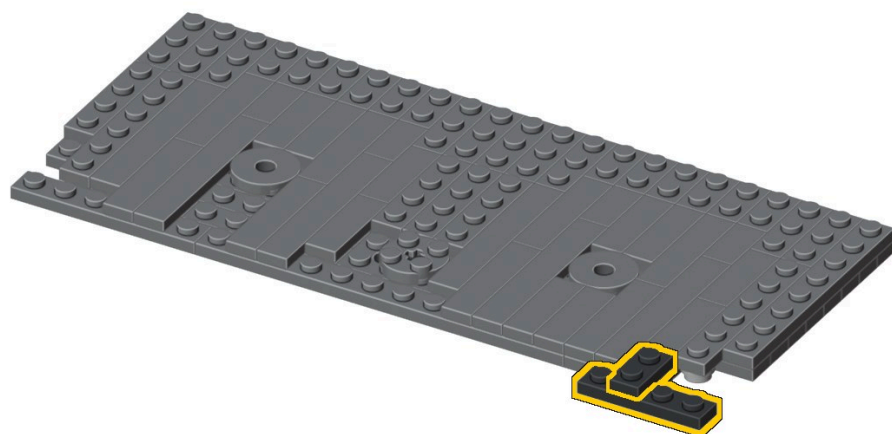
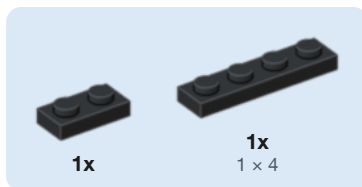


# 36

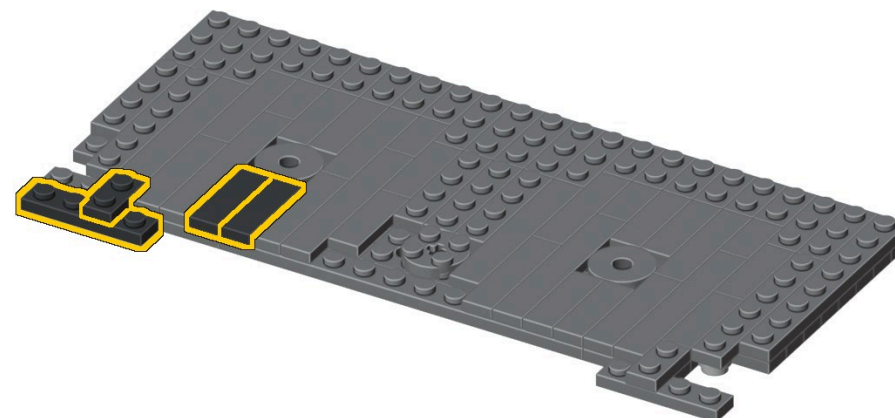
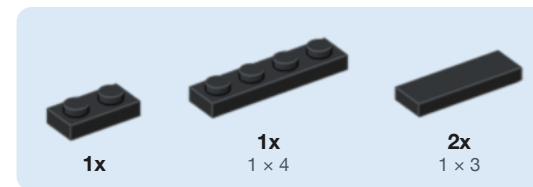
Chassis



# 37

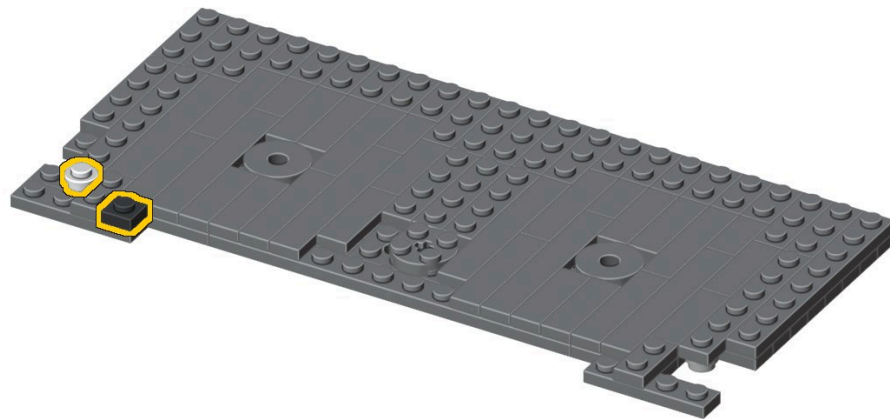
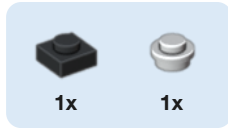


# 38

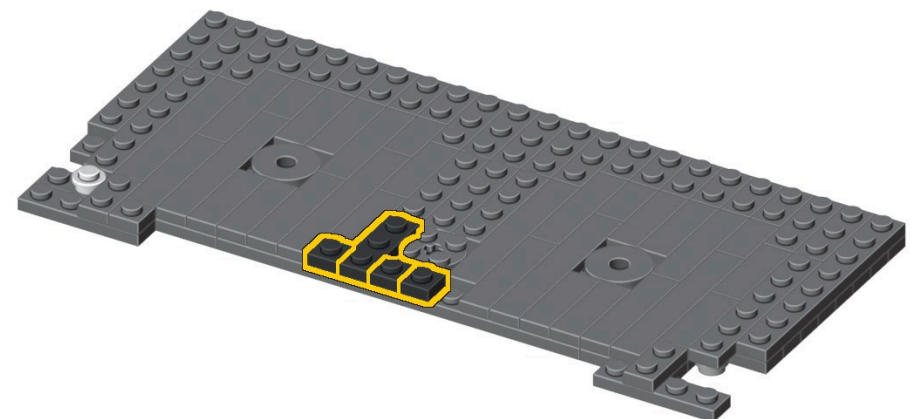
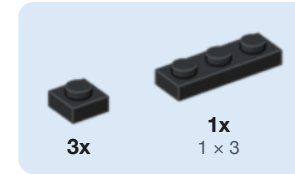


Chassis

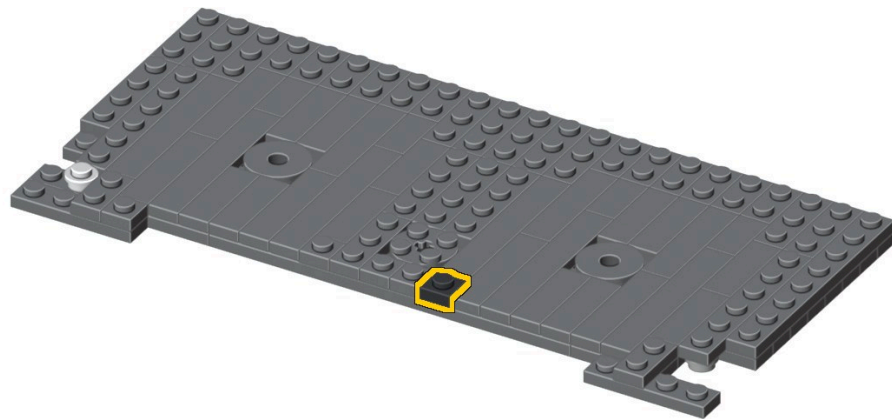
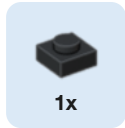
# 39



# 40

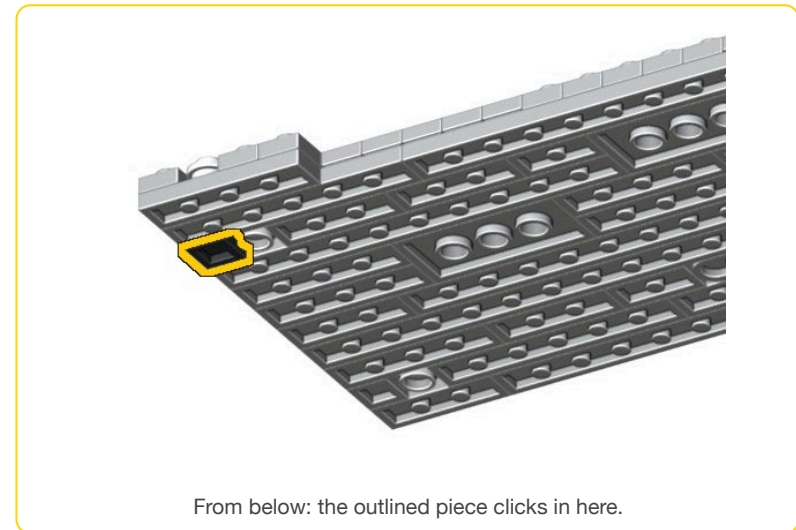
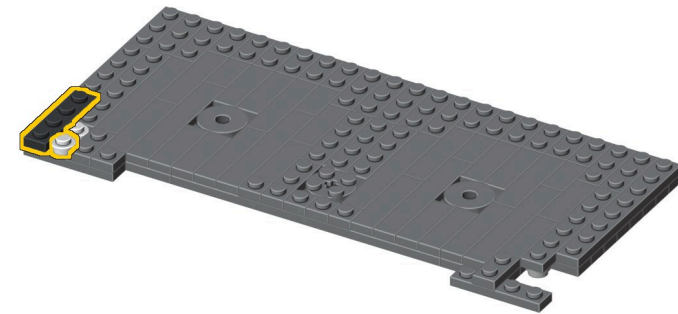
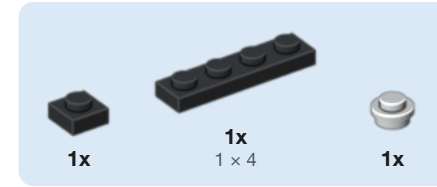


# 41

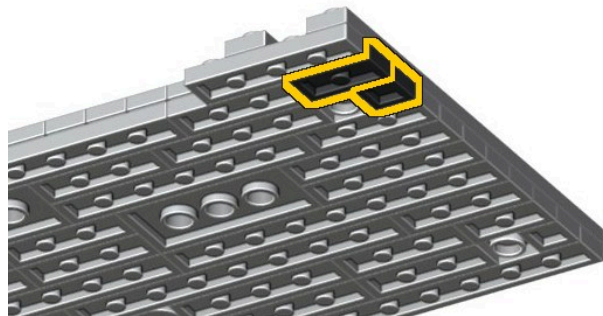
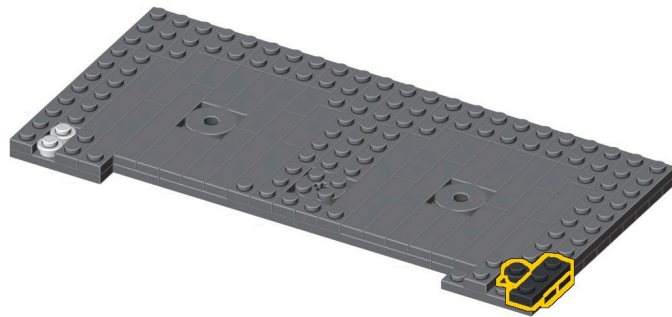
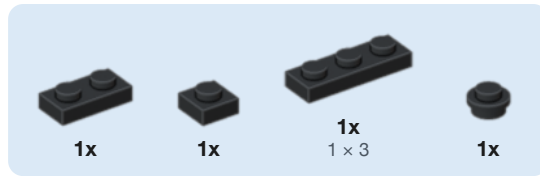


# 42

Chassis

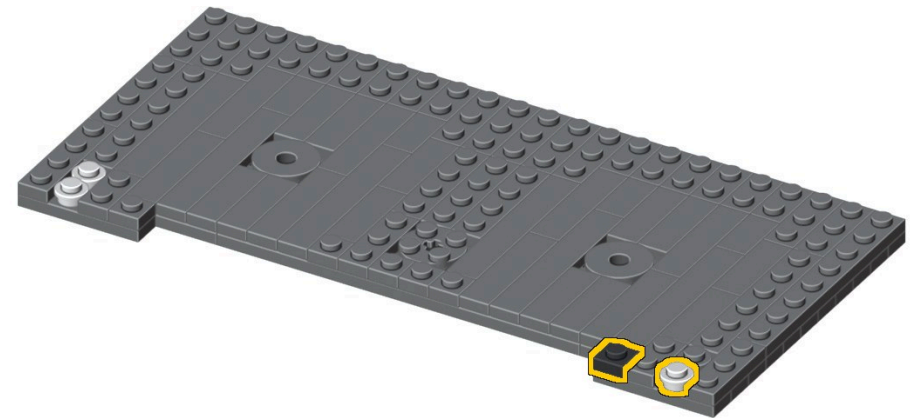
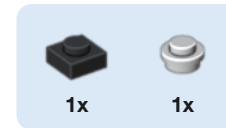


# 43



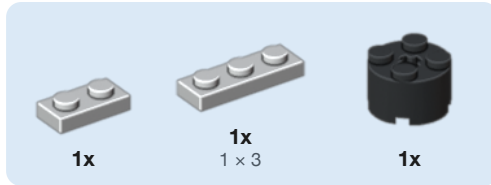
From below: the outlined pieces click in here.

# 44

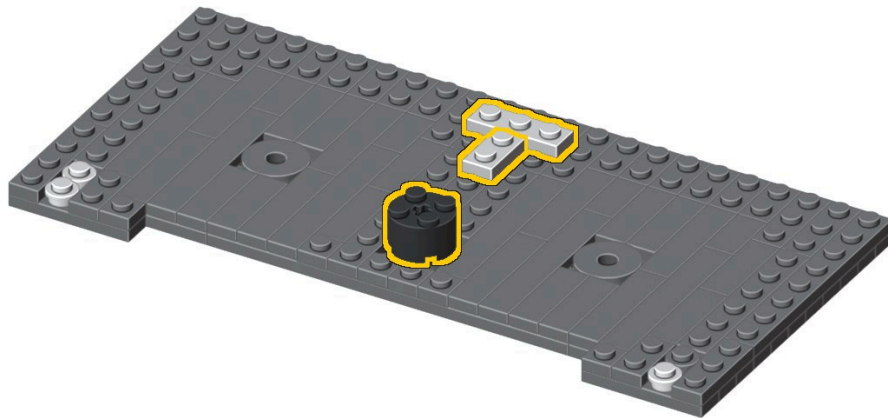


Chassis

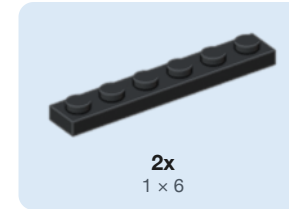
# 45



The lamp tube: a black 2 x 2 round brick over the lamp hole



# 46



Ledges the window hangers land on

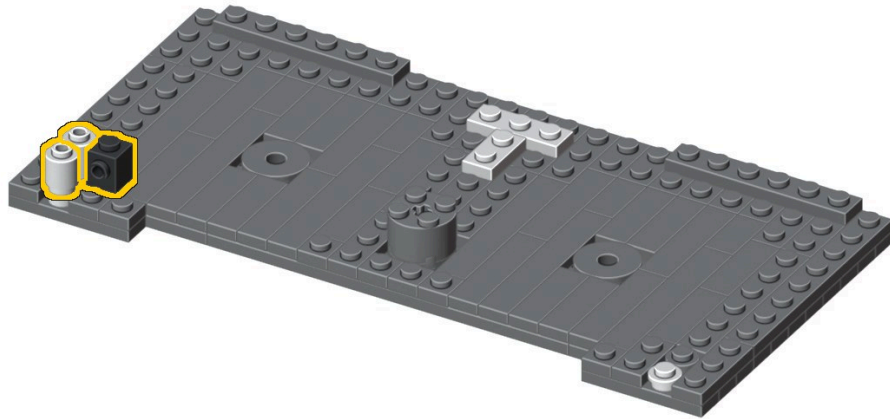


Chassis

# 47



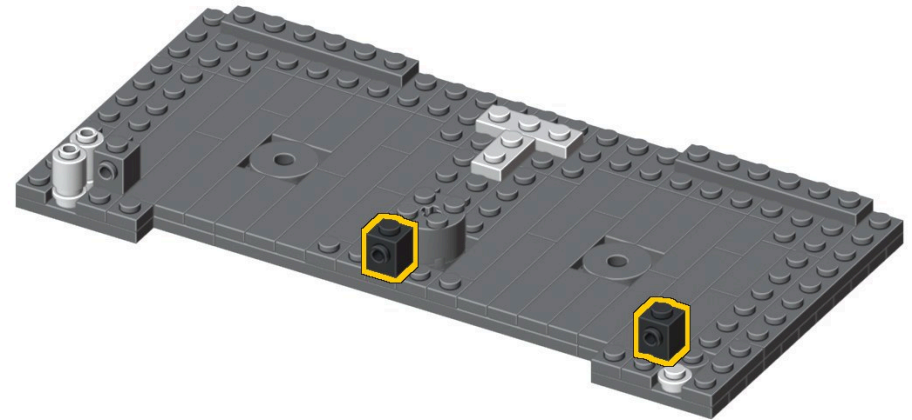
White tape guide rollers at the front corners, a white guide post and the posts the top shell's flap pivots land on



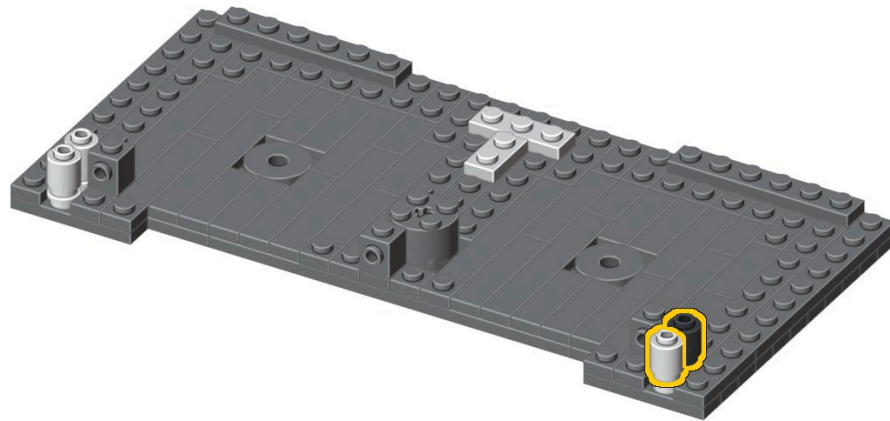
# 48



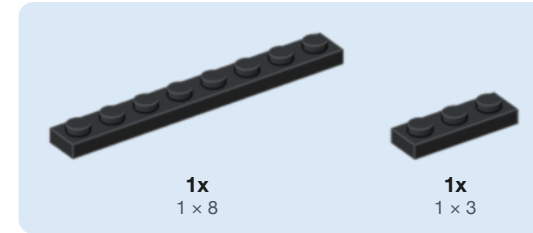
Chassis



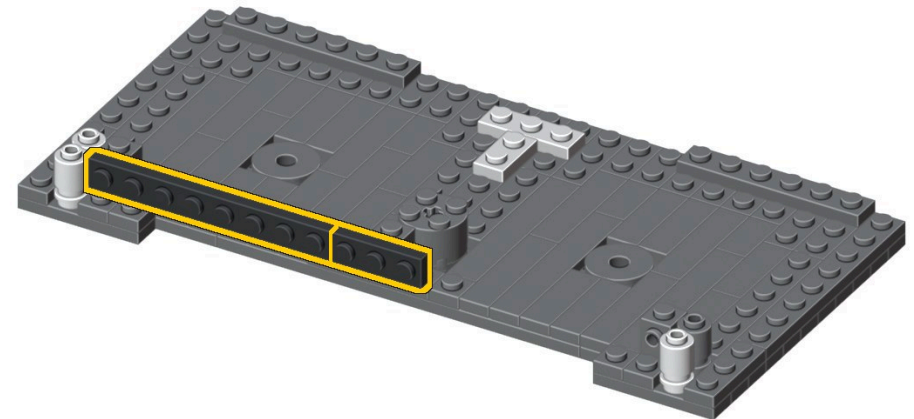
# 49



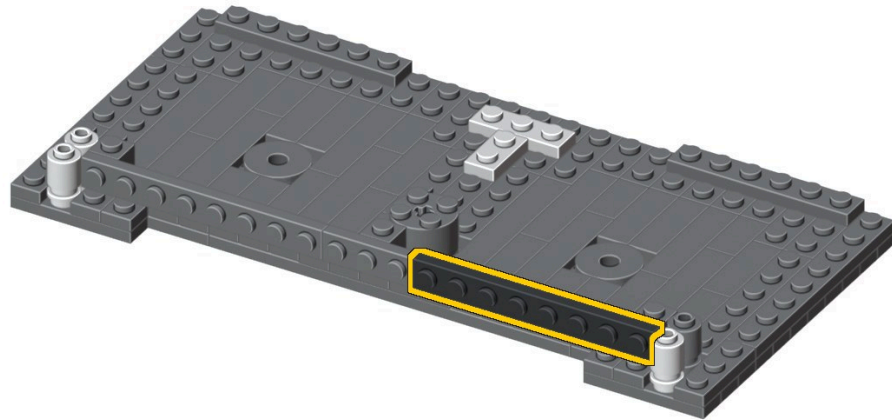
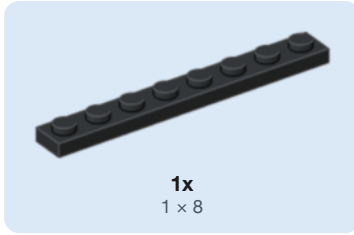
# 50



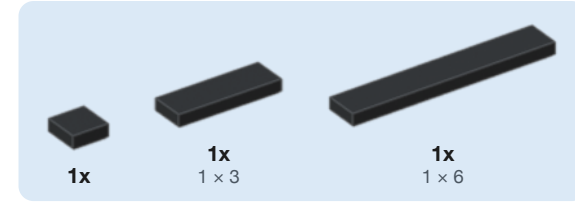
The tape run: a black 1 x N plate stood on its edge



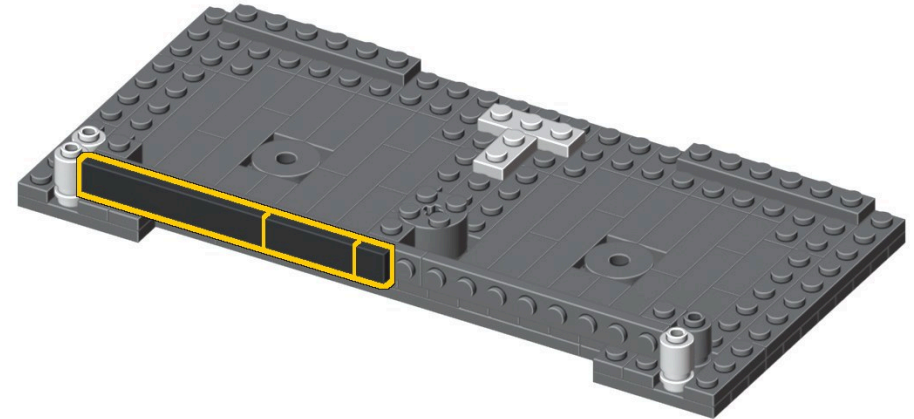
# 51



# 52

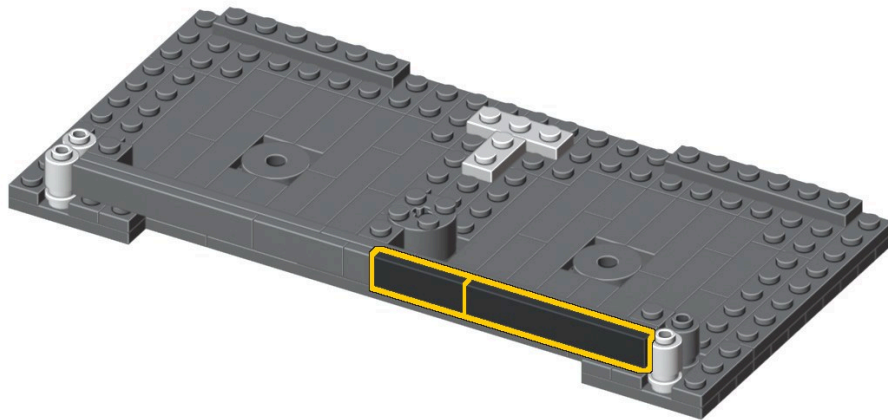
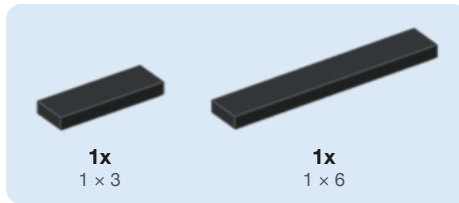


Black tiles on it: the tape stretched across the front



Chassis

# 53



SUPPLY REEL (FULL)

# 54

Supply reel (full)

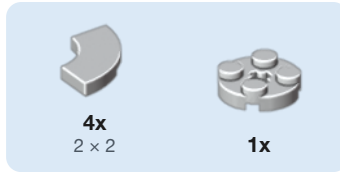


The lower flange and the tape pack: two black 8 x 8 round plates with a hole in the middle, one on the other

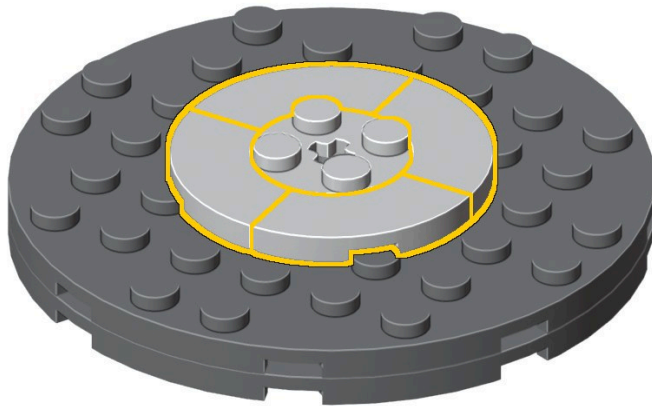


From below: the outlined piece clicks in here.

# 55



The hub: a round plate with an axle hole, and a flat white ring round it



# 56

Supply reel (full)



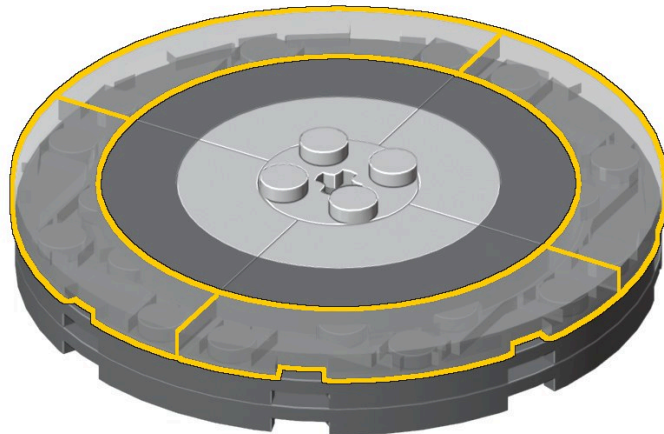
Tape wound round the hub



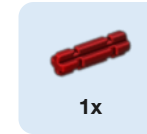
# 57



Clear upper flange



# 58



Supply reel (full)

seen from below

The red spindle: push the axle up through the hub



# VHS CASSETTE

## 59



Take the chassis and drop the supply reel onto its right-hand bearing: the reel's red axle goes down through the round holes



From below: the outlined piece clicks in here.

# TAKE-UP REEL

## 60

Take-up reel

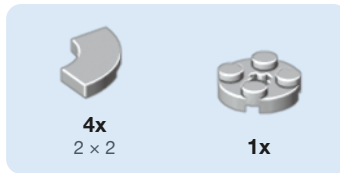


The lower flange and the tape pack: two black 8 x 8 round plates with a hole in the middle, one on the other

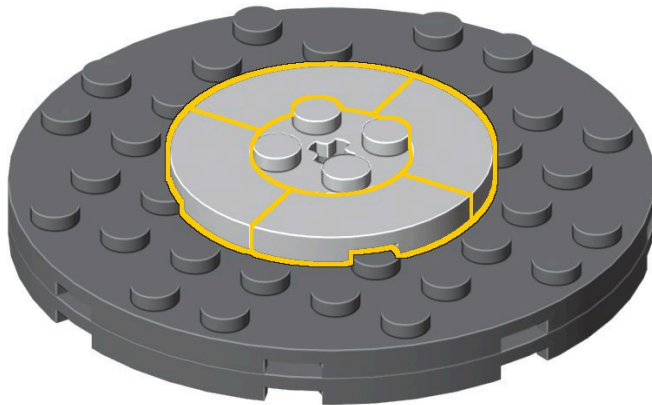


From below: the outlined piece clicks in here.

# 61



The hub: a round plate with an axle hole, and a flat white ring round it



# 62

Take-up reel



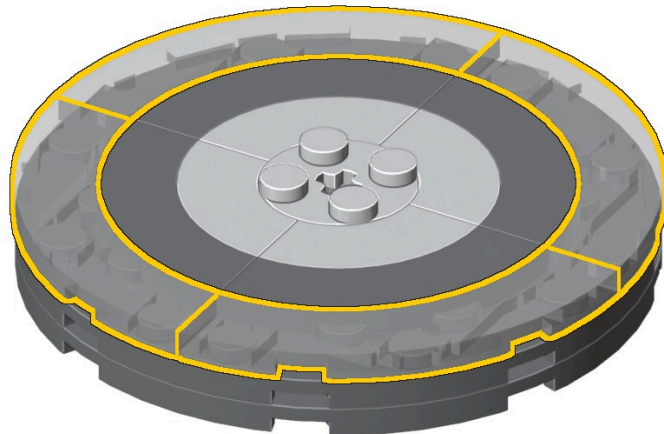
Just a few turns of tape left on the hub



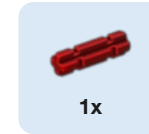
# 63



Clear upper flange



# 64



The red spindle: push the axle up through the hub

Take-up reel

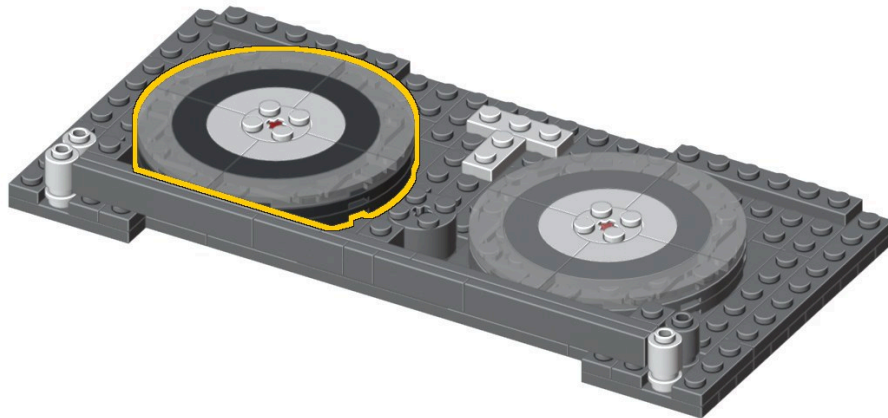
seen from below



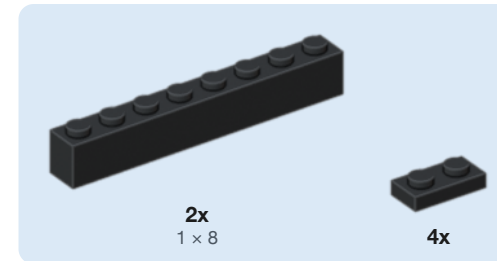
65



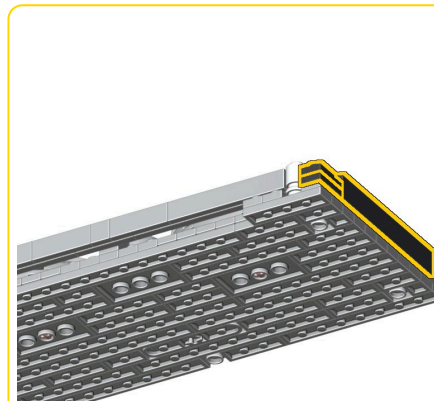
The take-up reel on the left-hand bearing



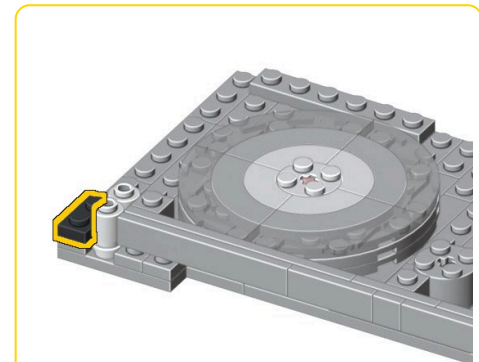
66



End walls



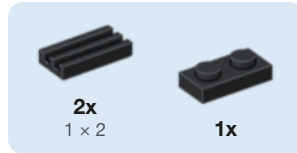
From below: the outlined pieces click in here.



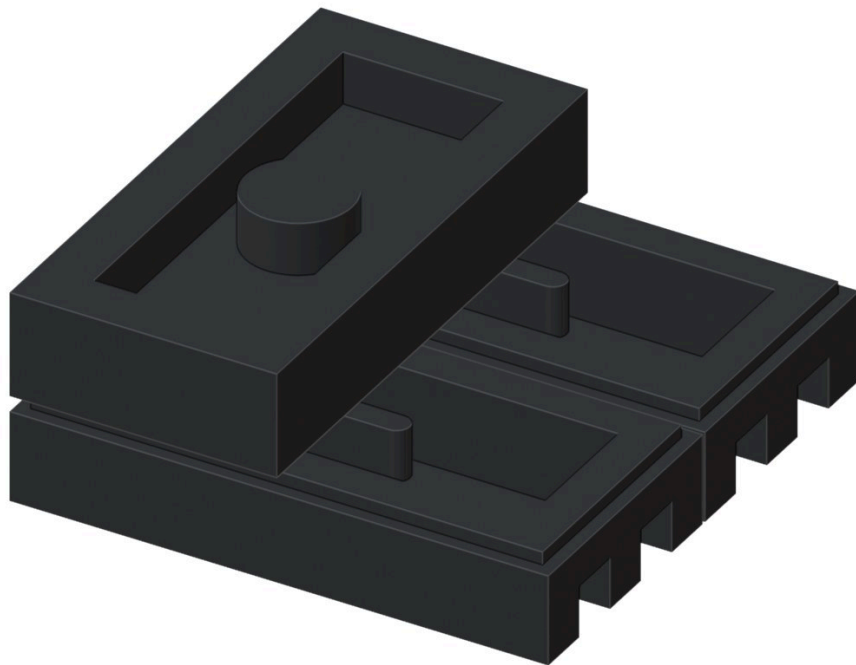
First the outlined piece: the others go on over it.

# TOP SHELL

# 67

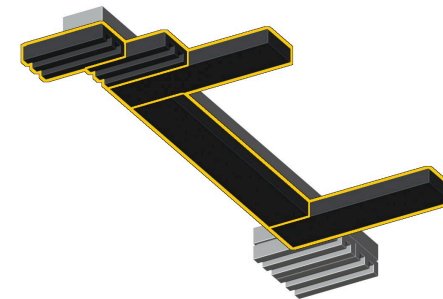
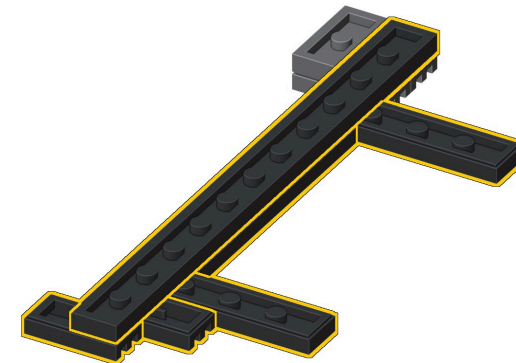
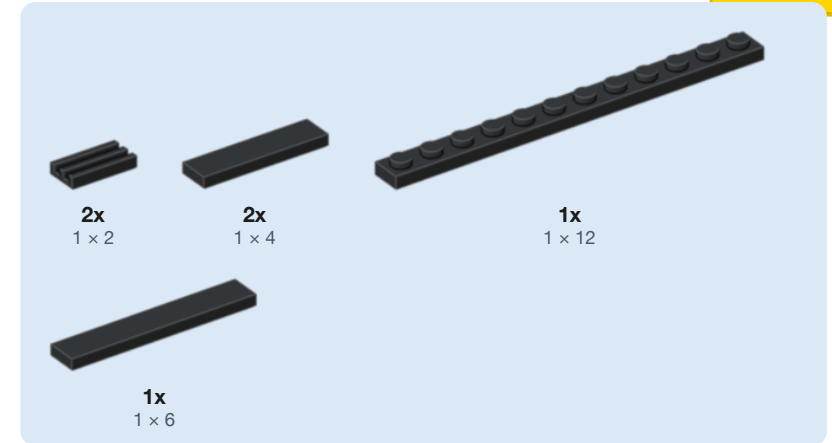


Top shell, built upside down: the grip-band grilles and the black rims go face down, then plates on them



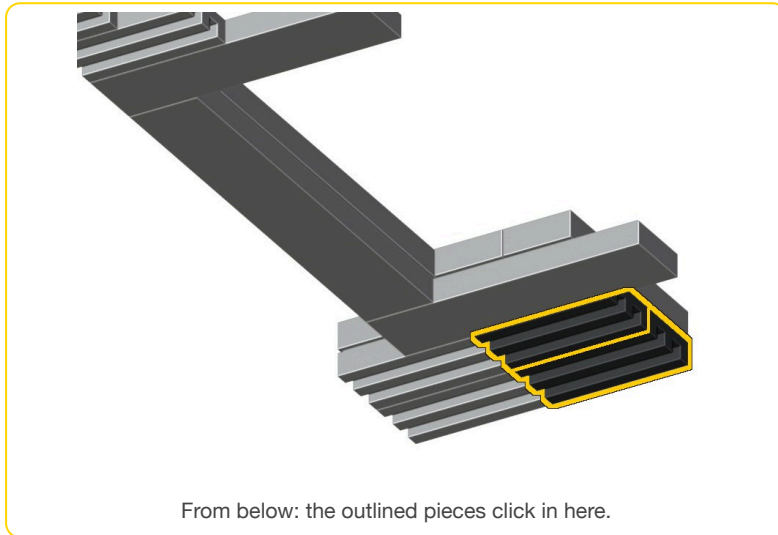
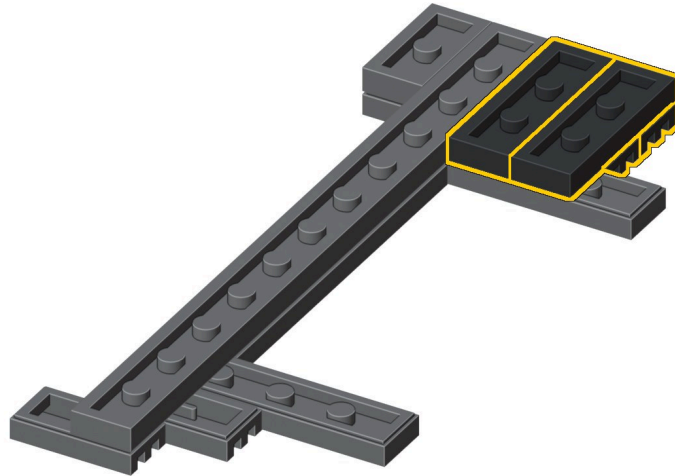
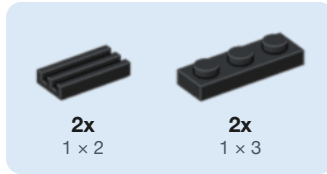
# 68

# Top shell

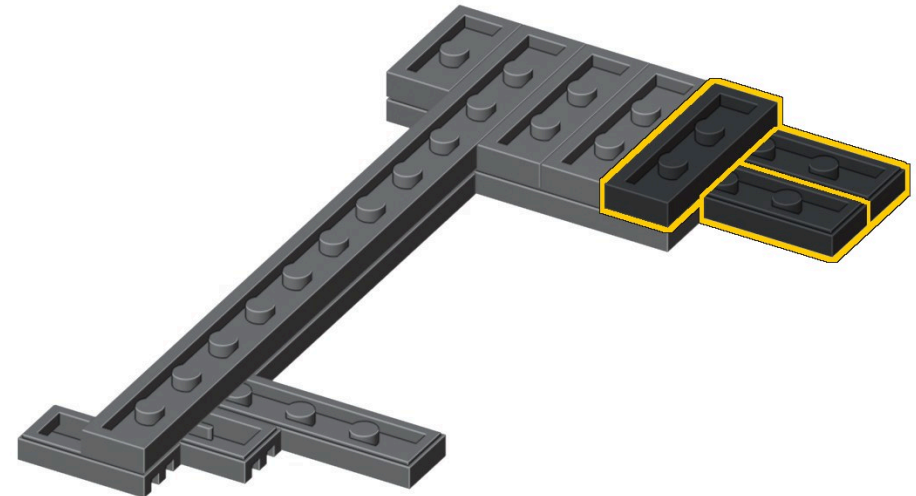
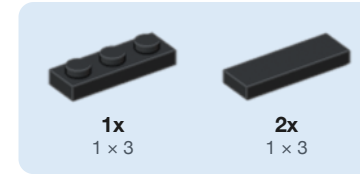


From below: the outlined pieces click in here.

# 69

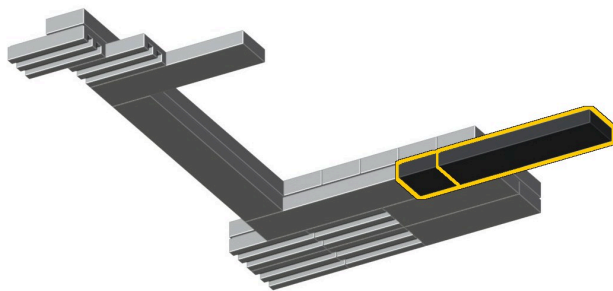
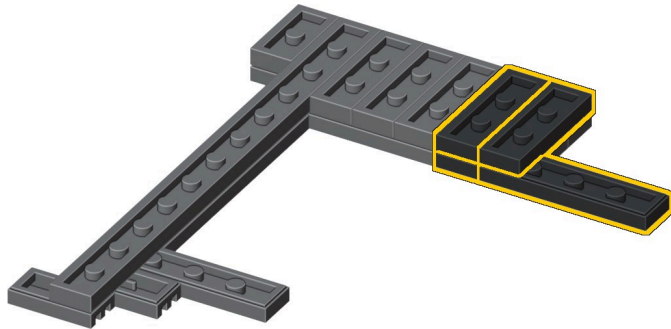
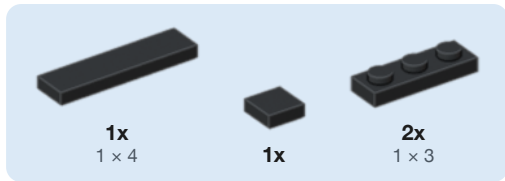


# 70



Top shell

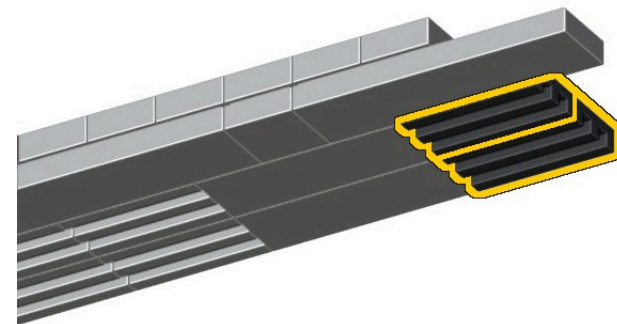
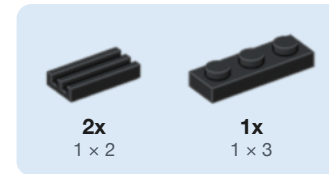
# 71



From below: the outlined pieces click in here.

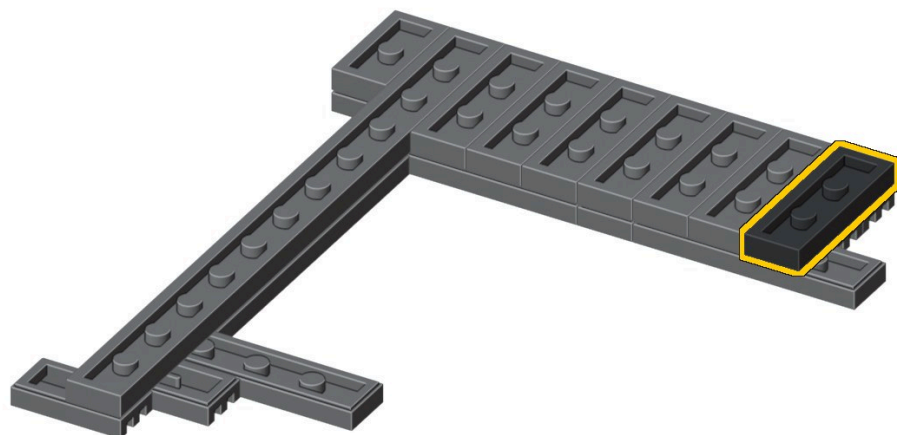
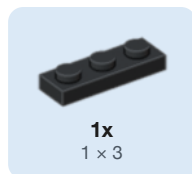
# 72

Top shell



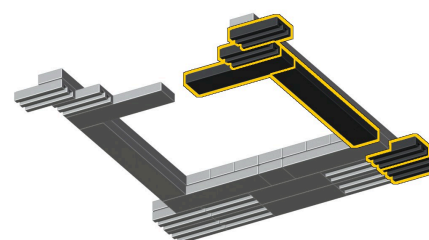
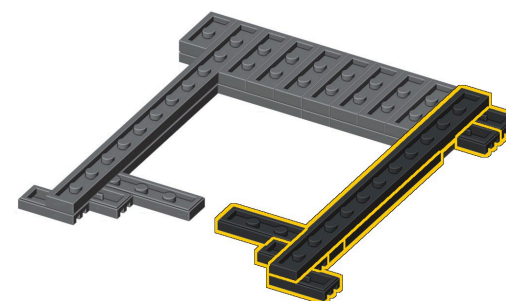
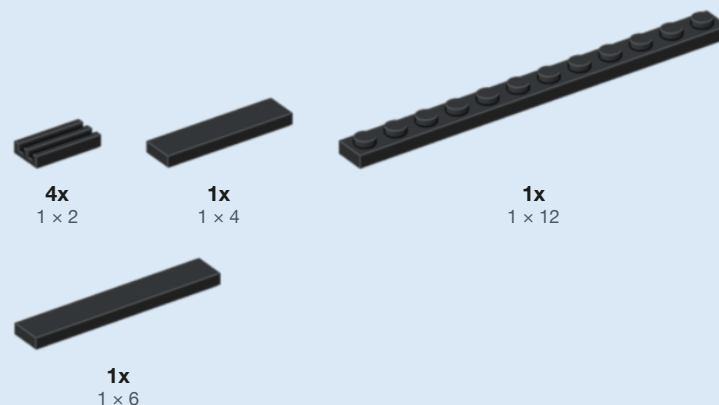
From below: the outlined pieces click in here.

# 73



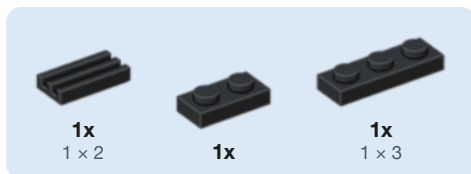
# 74

Top shell

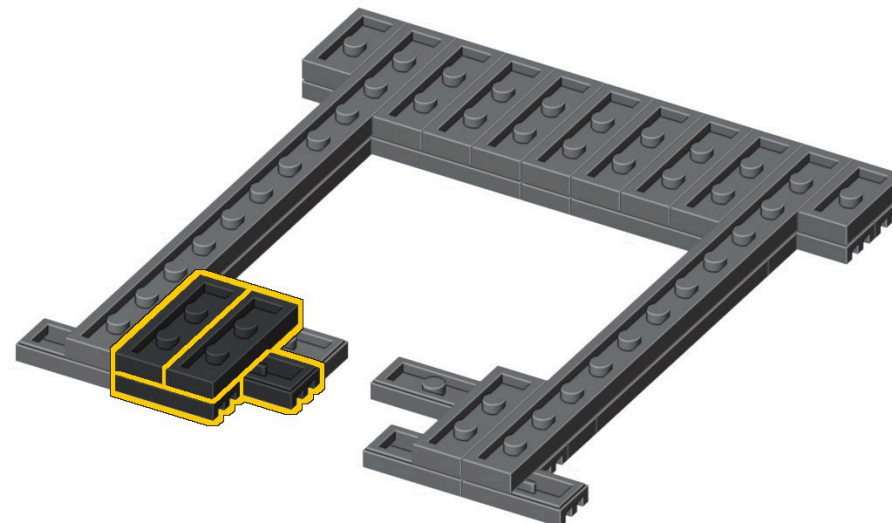
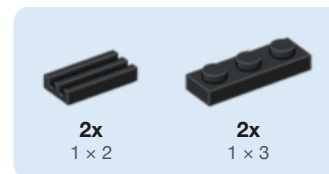


From below: the outlined pieces click in here.

# 75

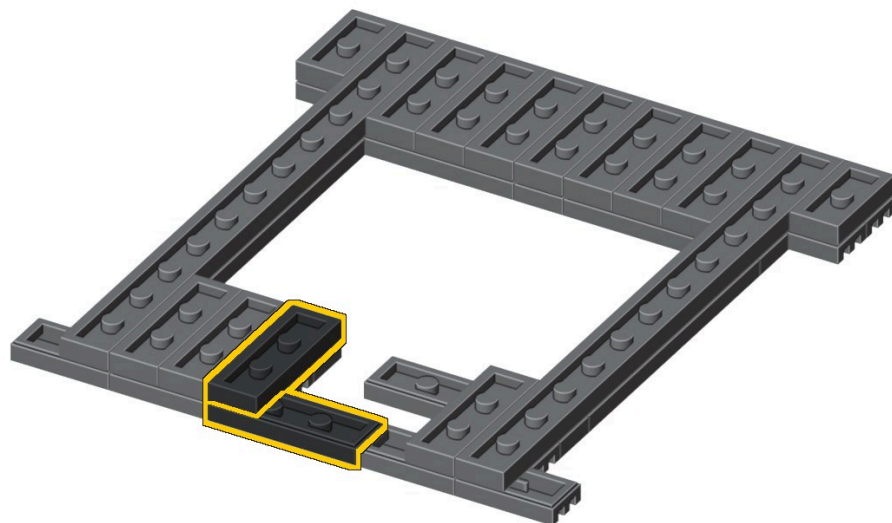
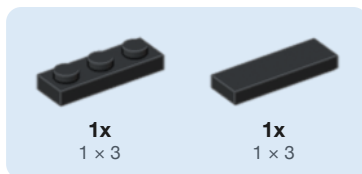


# 76

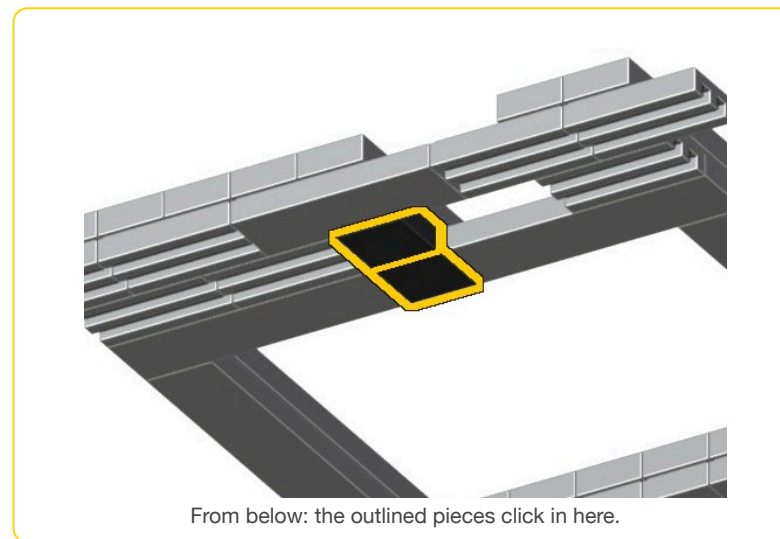
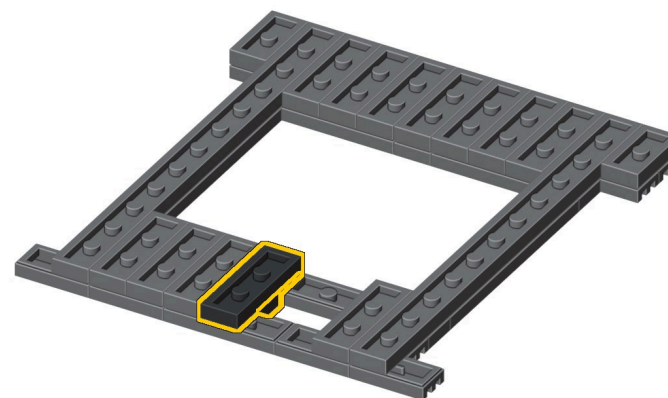
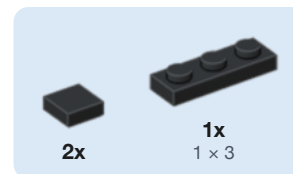


Top shell

# 77

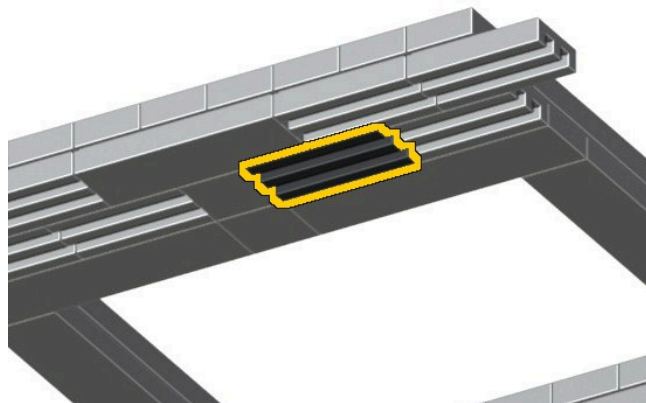
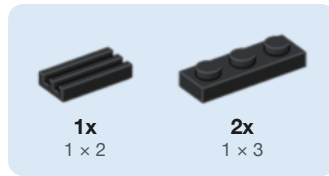


# 78



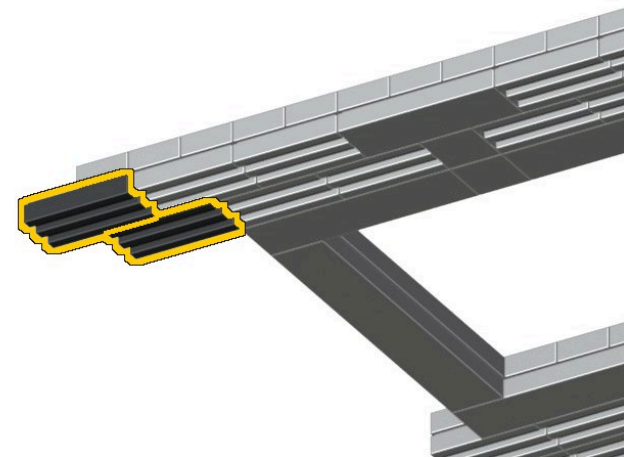
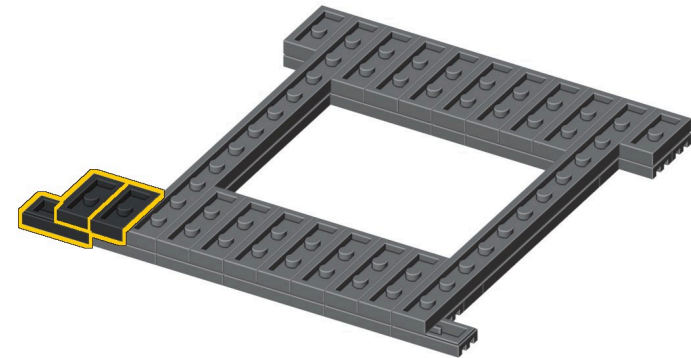
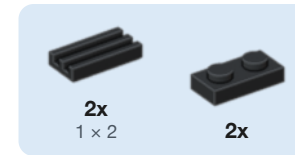
Top shell

# 79



From below: the outlined piece clicks in here.

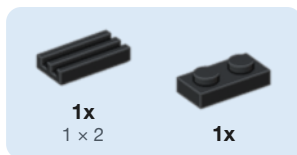
# 80



From below: the outlined pieces click in here.

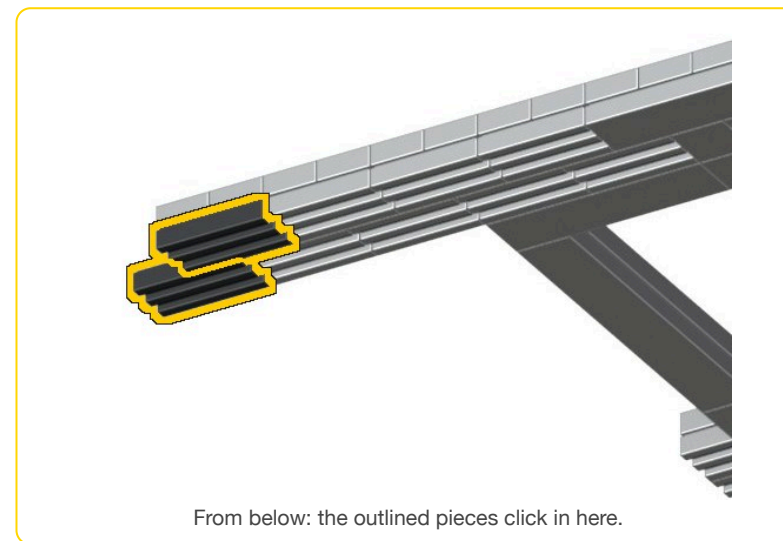
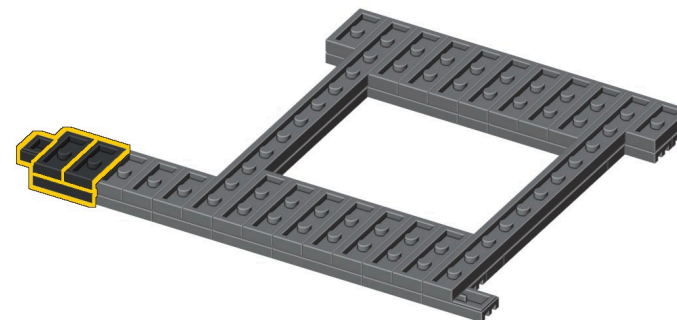
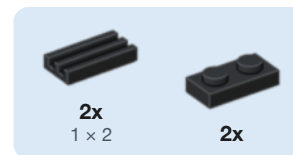
Top shell

# 81

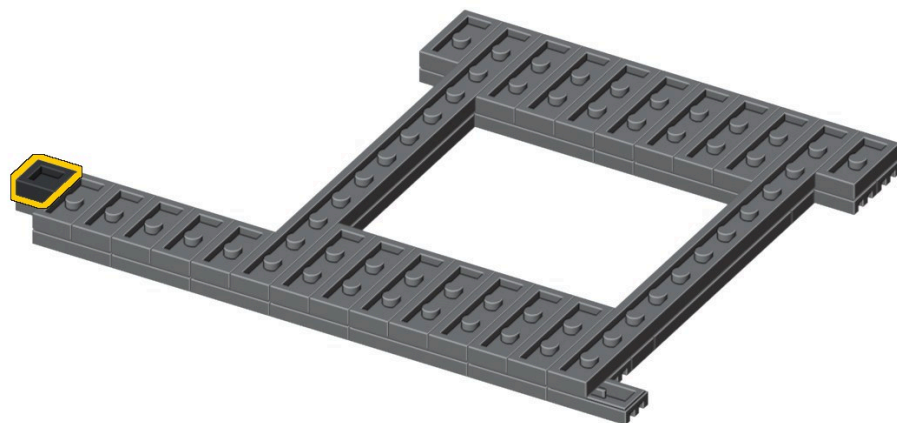


# 82

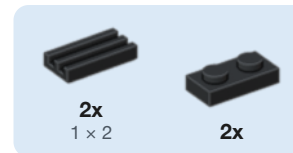
Top shell



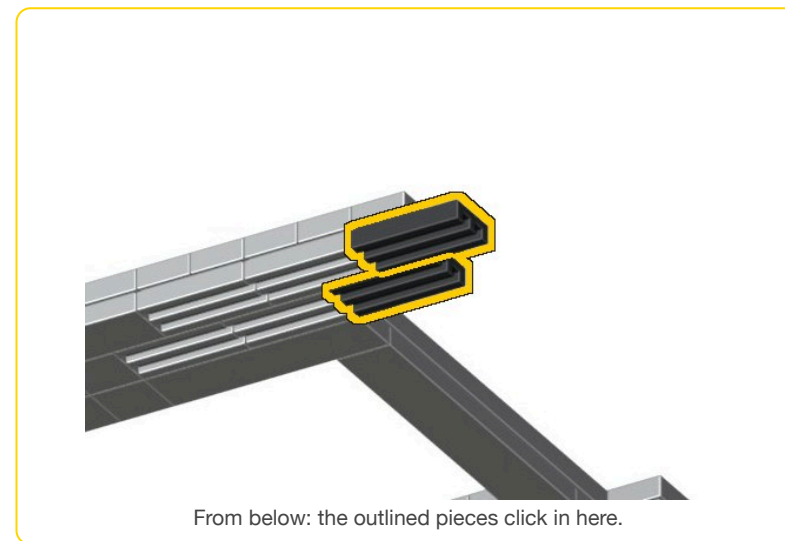
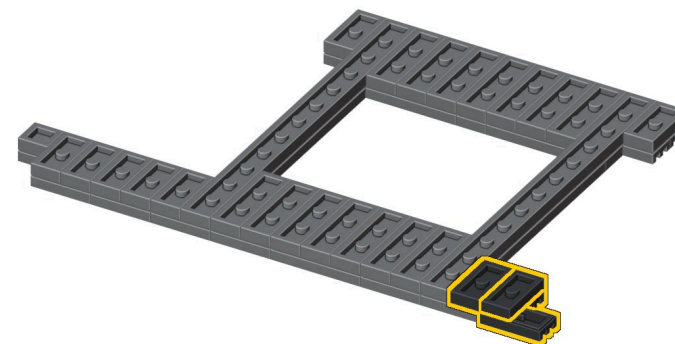
# 83



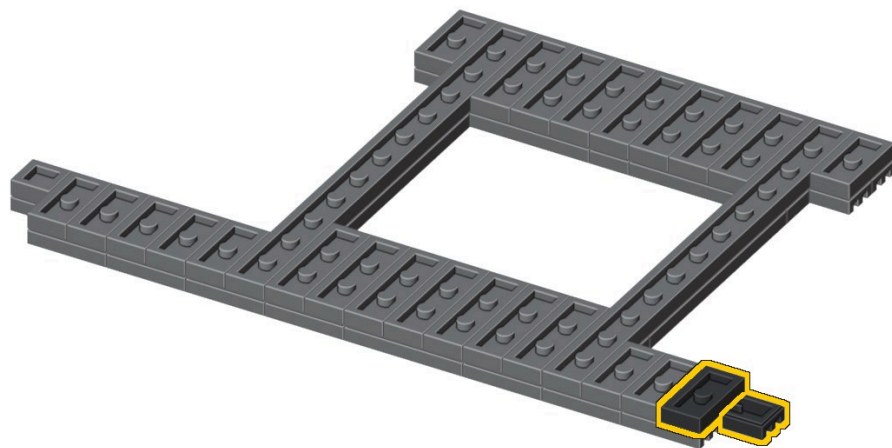
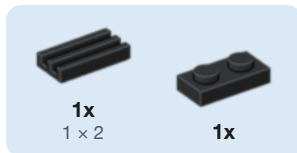
# 84



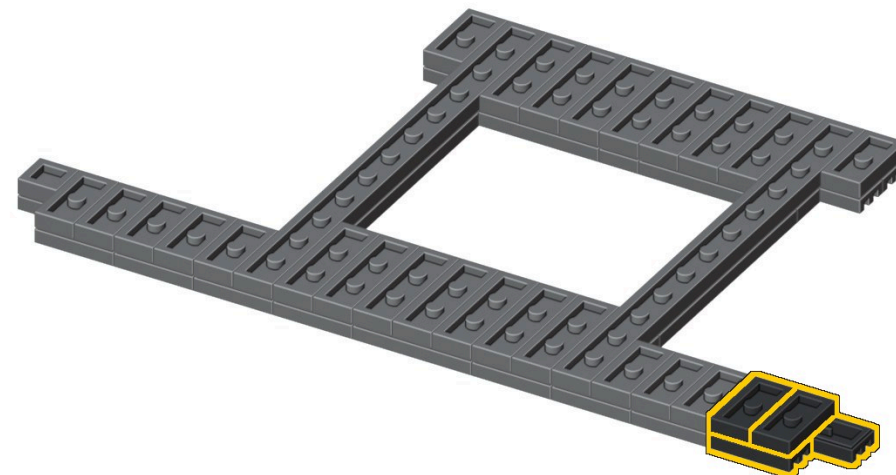
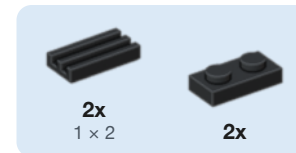
Top shell



# 85

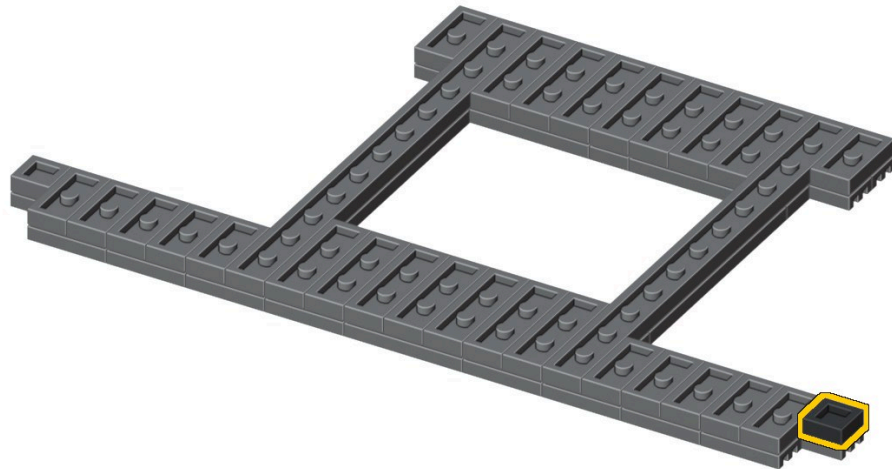
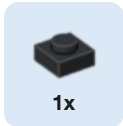


# 86



Top shell

87

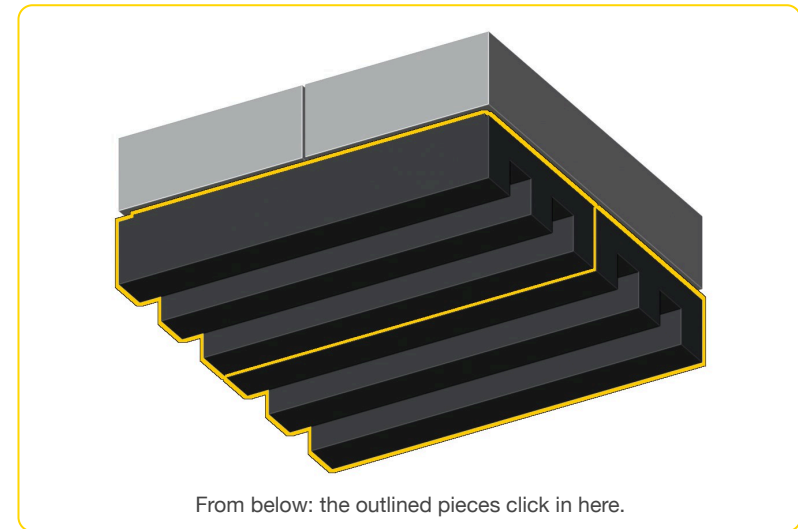
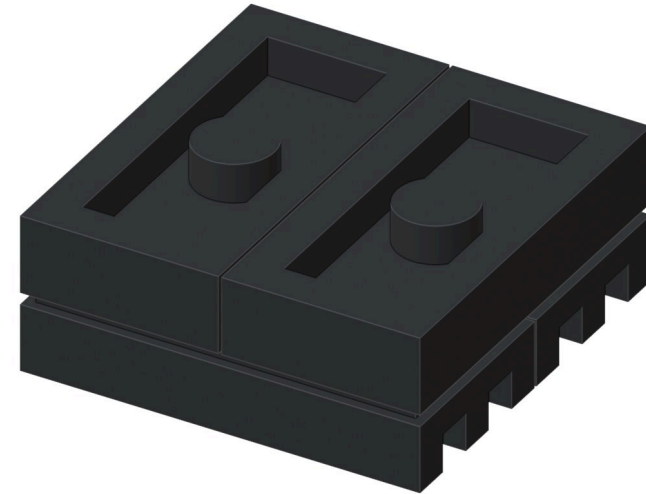


TOP SHELL: STRIP 1

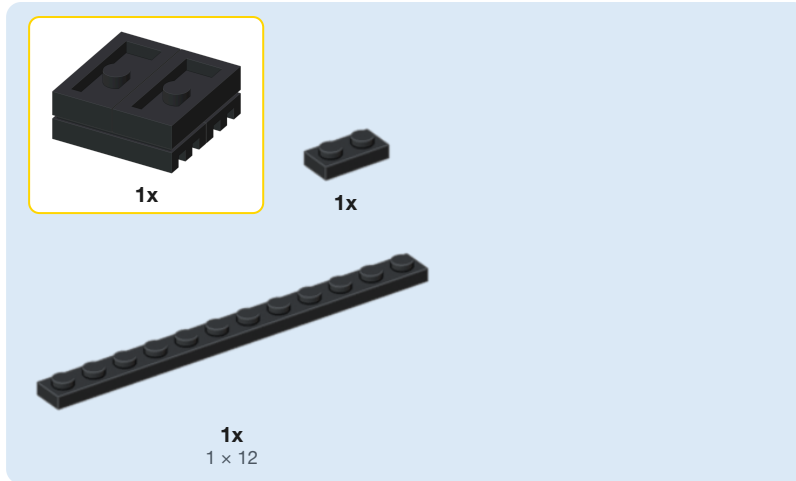
88



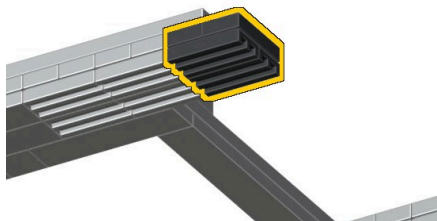
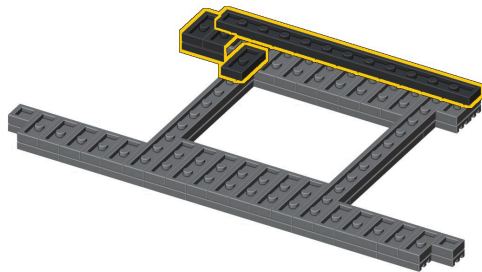
Build this strip apart, the way up it is drawn. Once built, it goes on with the piece that locks it down



Top shell: strip 1



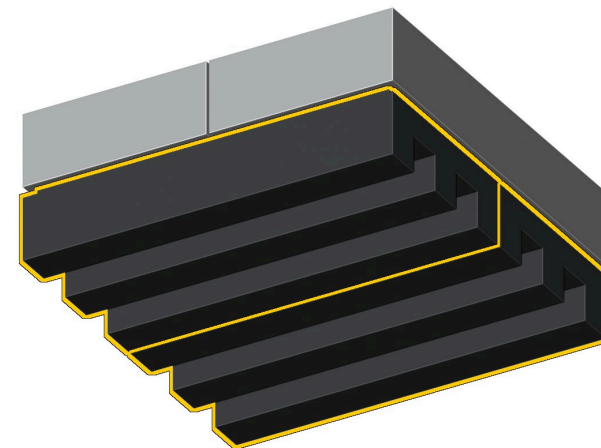
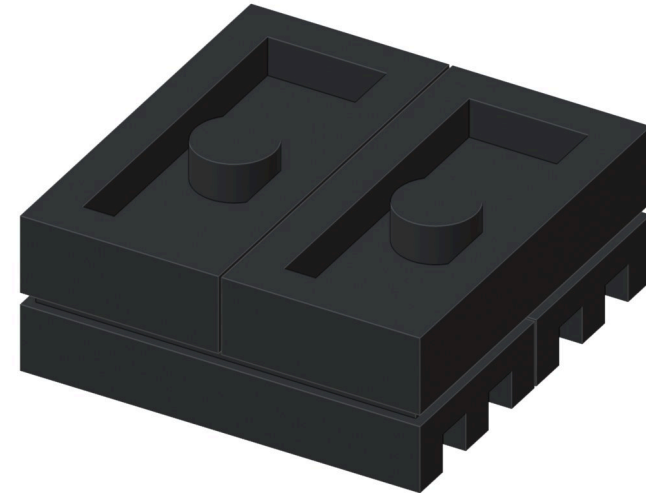
A third layer, with the flap's pivot blocks (plates with a pin hole)



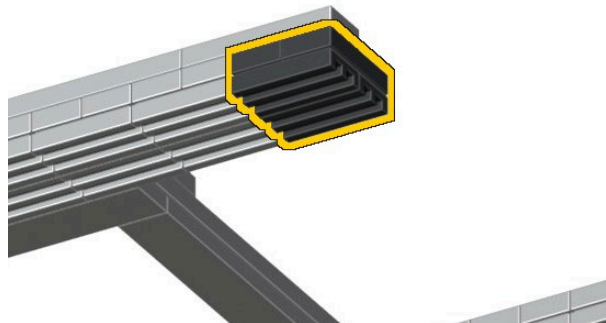
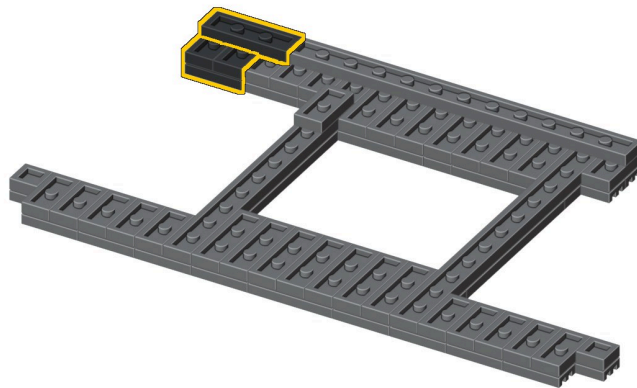
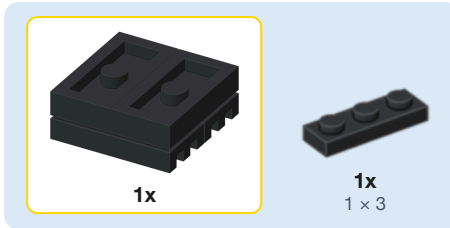
From behind and below: the outlined piece.



Build this strip apart, the way up it is drawn. Once built, it goes on with the piece that locks it down



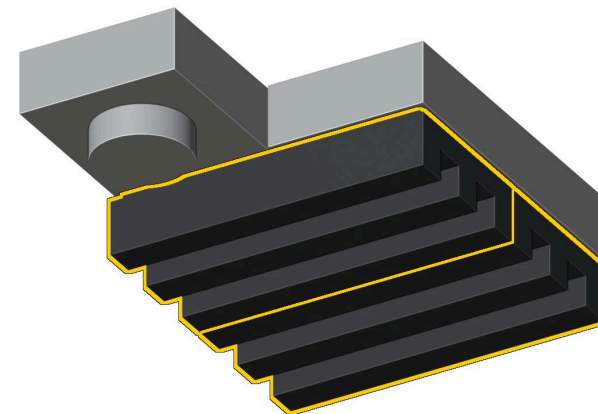
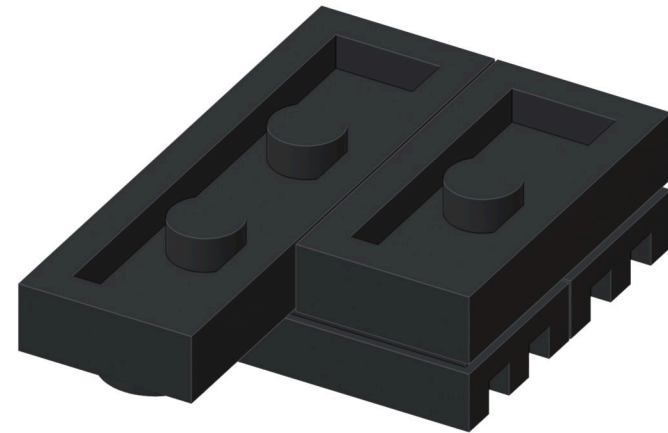
From below: the outlined pieces click in here.



From behind and below: the outlined piece.

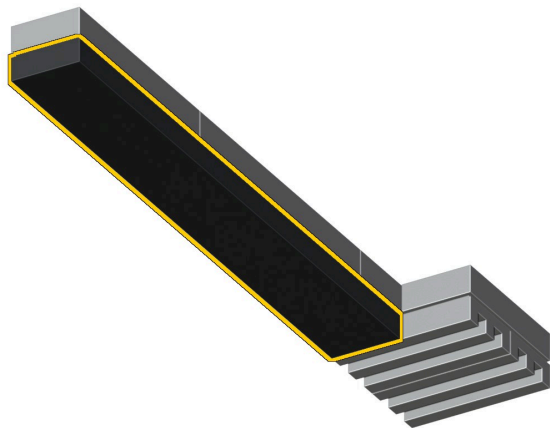
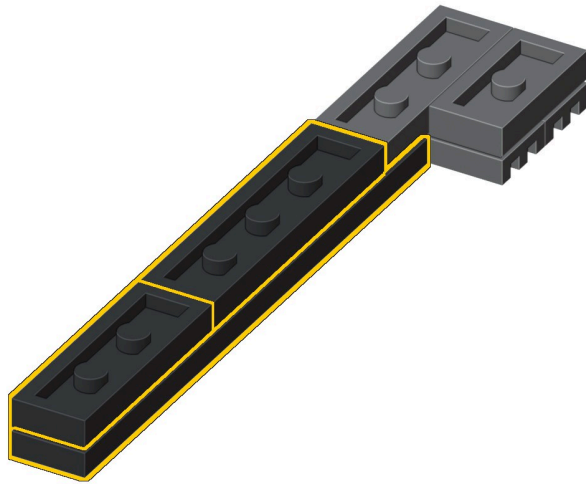
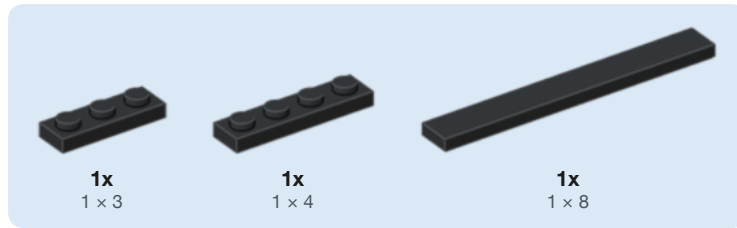


Build this strip apart, the way up it is drawn. Once built, it goes on with the piece that locks it down



From below: the outlined pieces click in here.

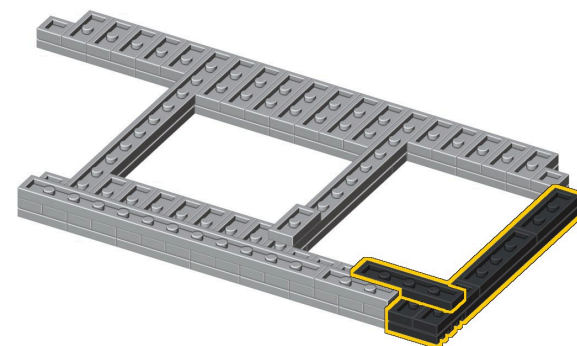
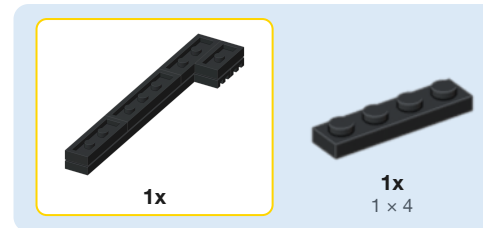
# 93



From below: the outlined piece clicks in here.

# 94

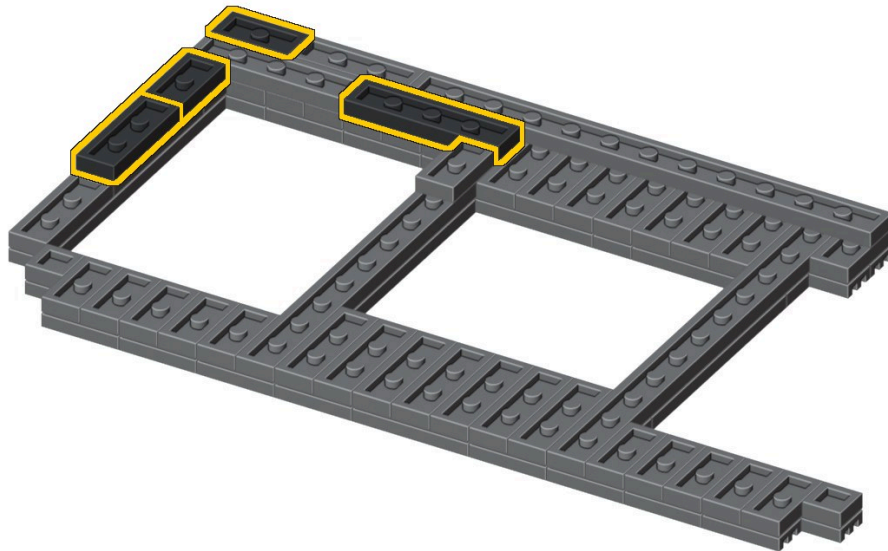
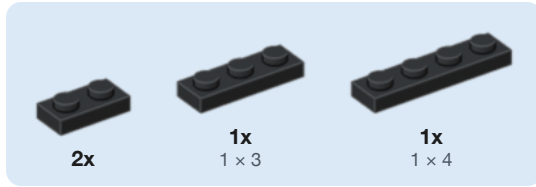
TOP SHELL



From behind: the outlined pieces.

Top shell

# 95

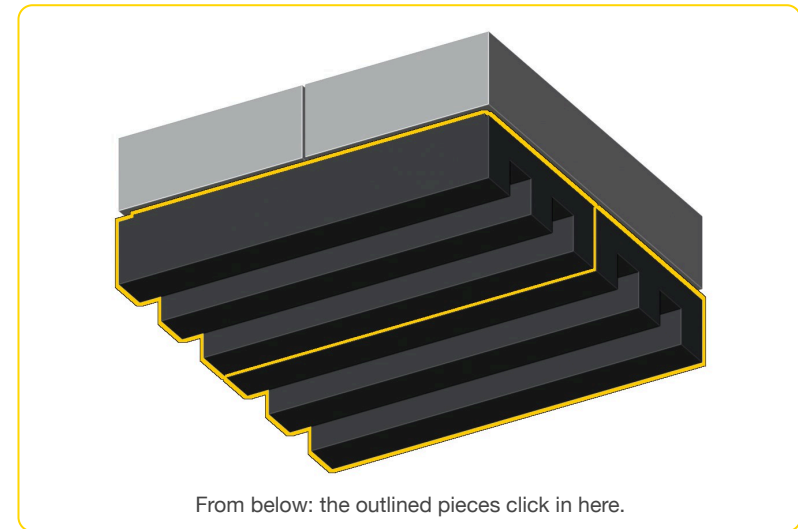
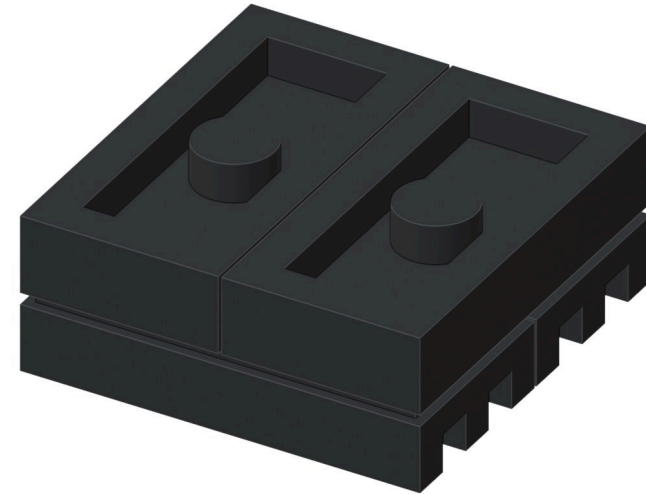


TOP SHELL: STRIP 4

# 96

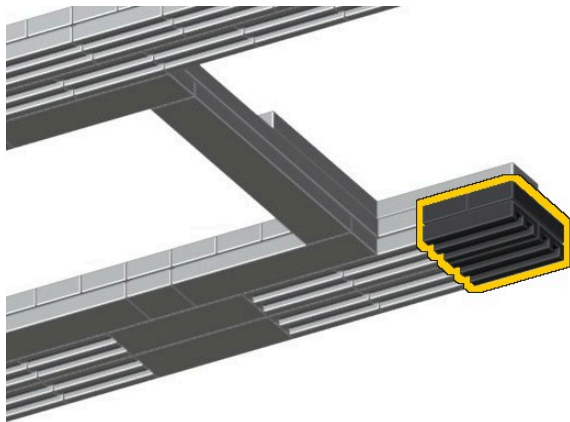
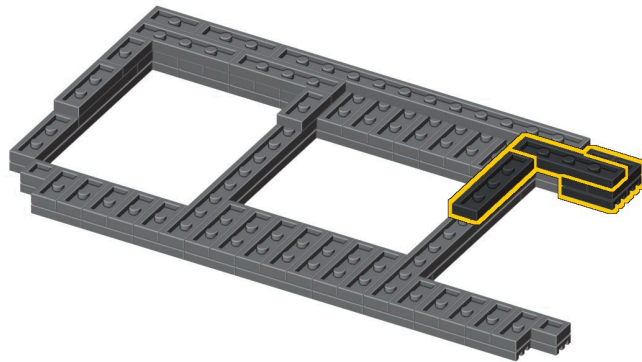
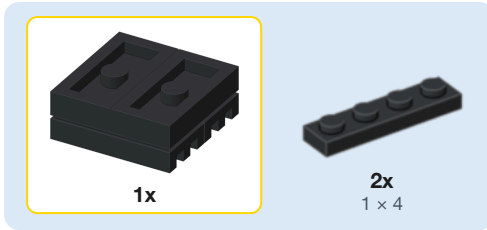


Build this strip apart, the way up it is drawn. Once built, it goes on with the piece that locks it down



TOP SHELL

97



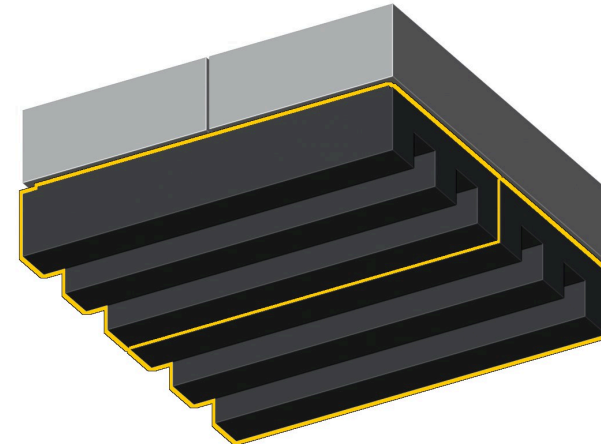
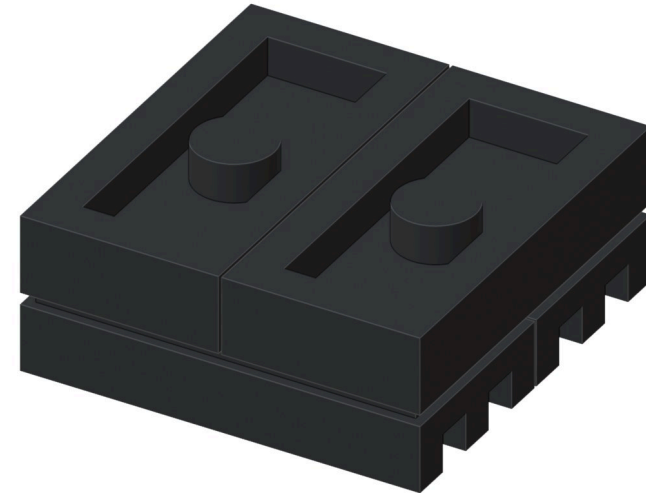
From below: the outlined piece clicks in here.

TOP SHELL: STRIP 5

98



Build this strip apart, the way up it is drawn. Once built, it goes on with the piece that locks it down

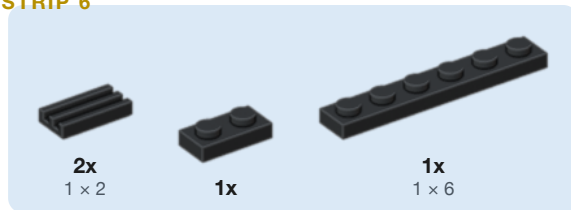


From below: the outlined pieces click in here.

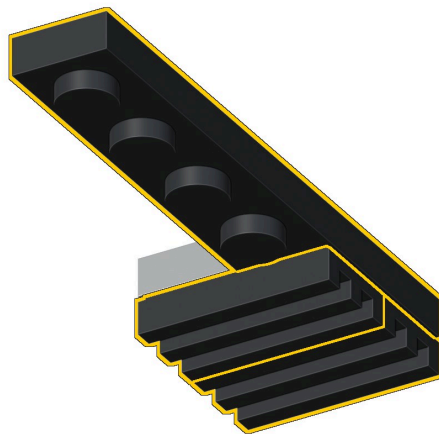
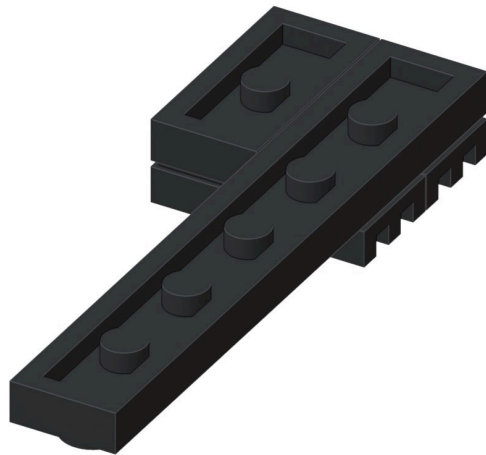
Top shell: strip 5

TOP SHELL: STRIP 6

99



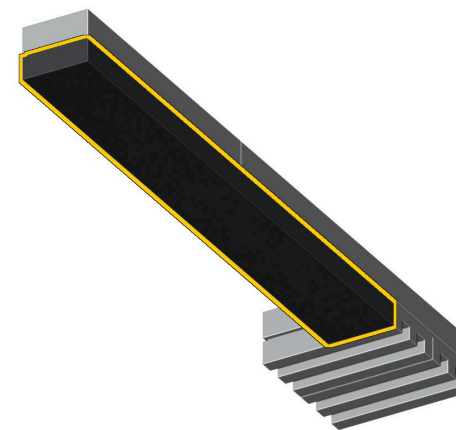
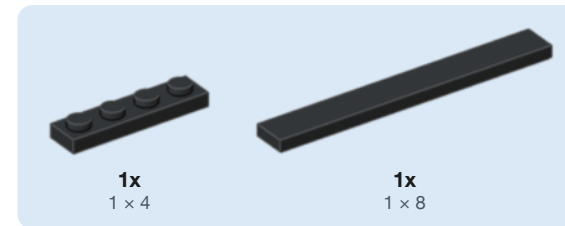
Build this strip apart, the way up it is drawn. Once built, it goes on with the piece that locks it down



From below: the outlined pieces click in here.

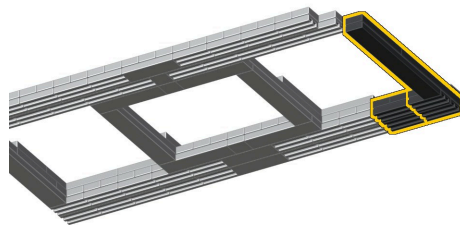
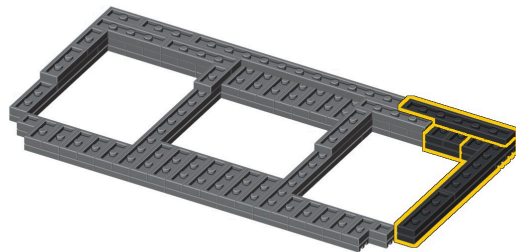
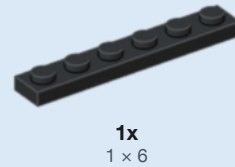
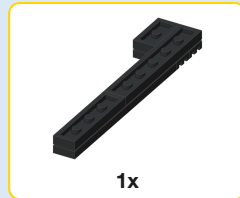
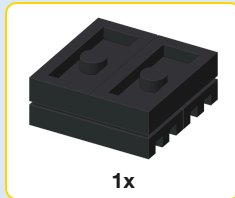
100

Top shell: strip 6



From below: the outlined piece clicks in here.

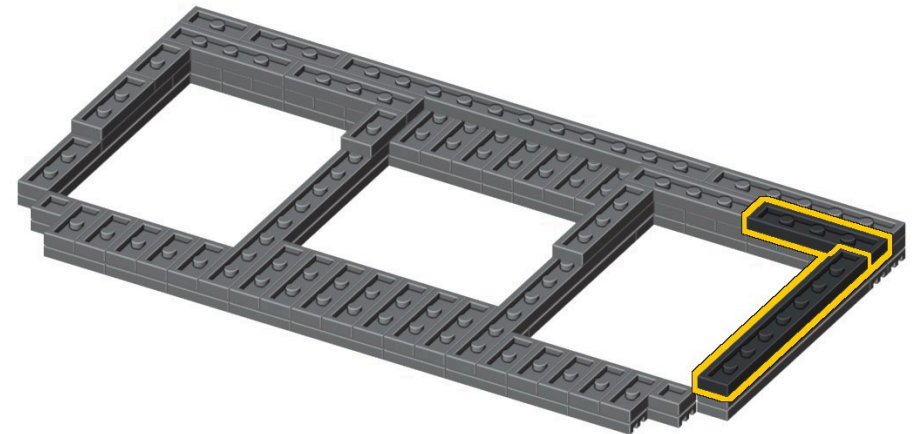
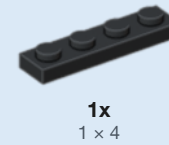
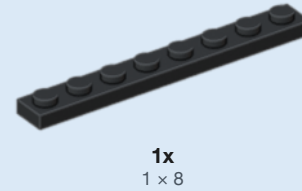
TOP SHELL  
101



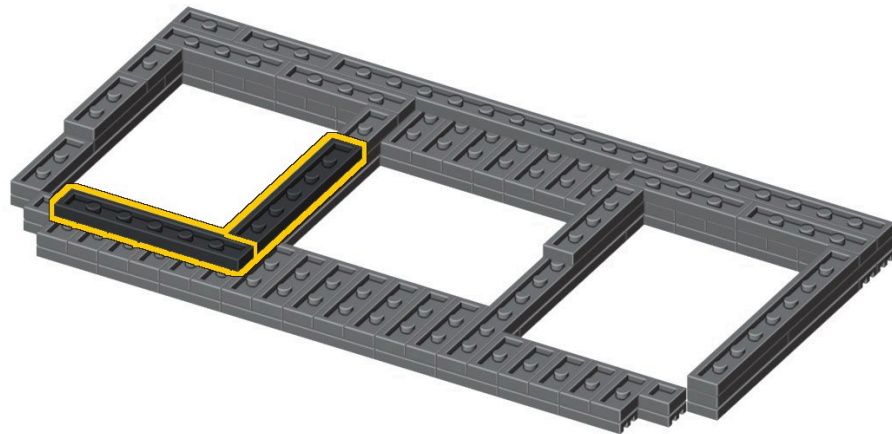
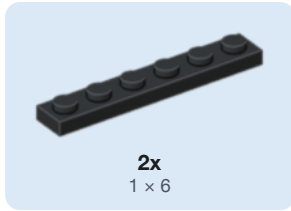
From below: the outlined pieces click in here.

102

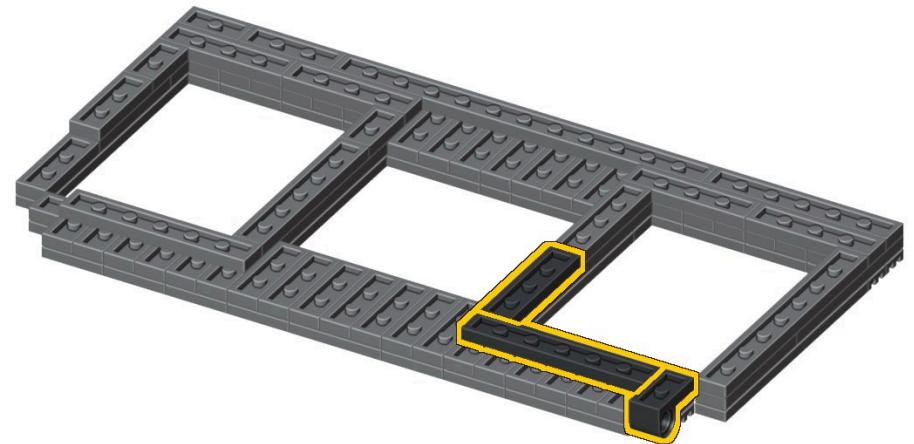
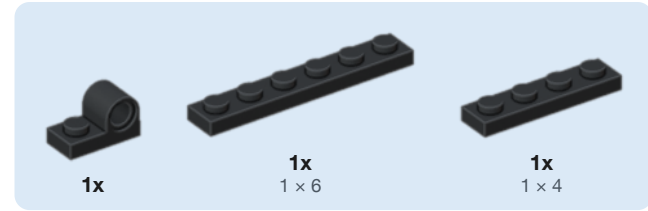
Top shell



# 103

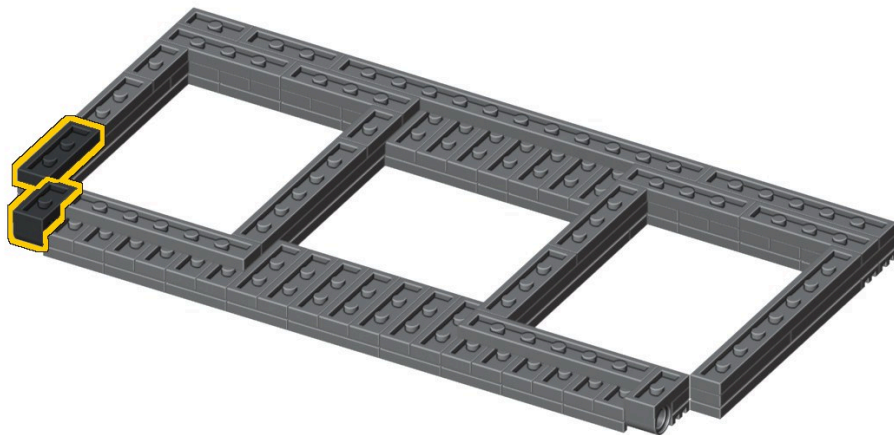
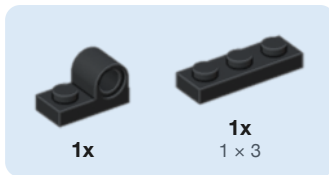


# 104



Top shell

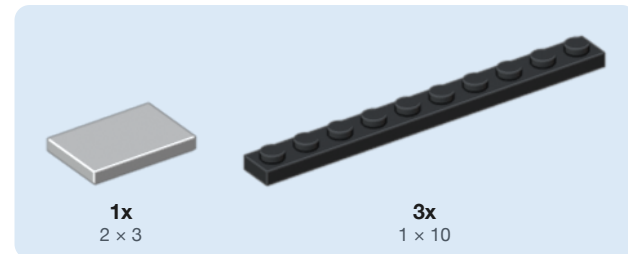
# 105



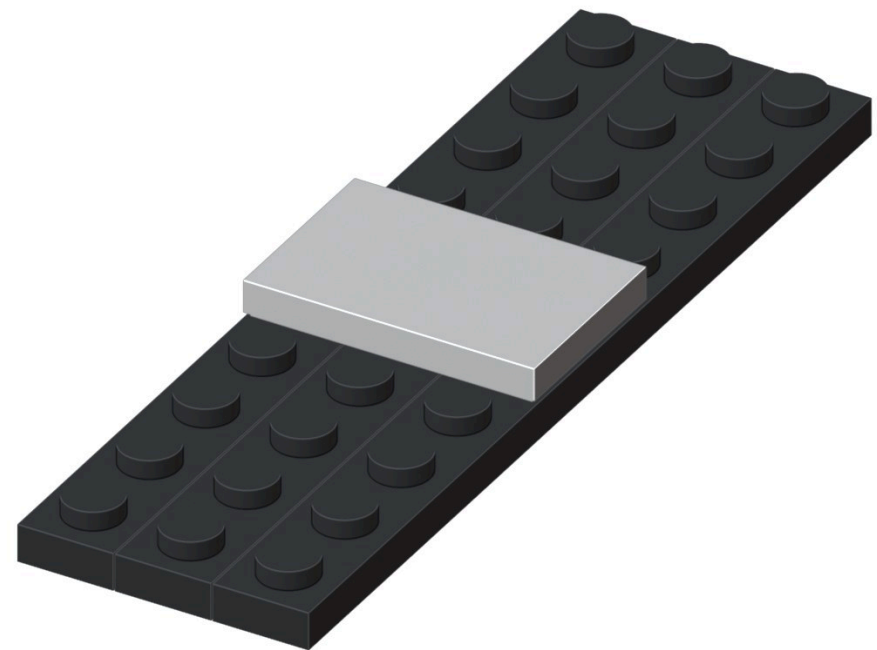
FACE LABEL

# 106

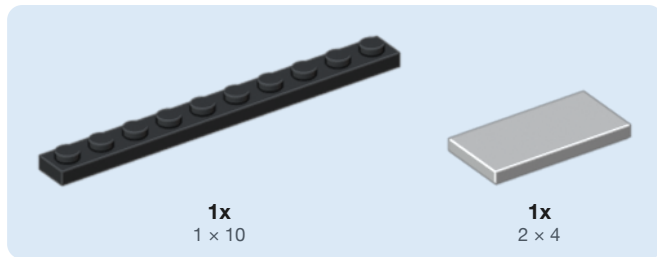
Face label



The face label: black plates running front to back ...

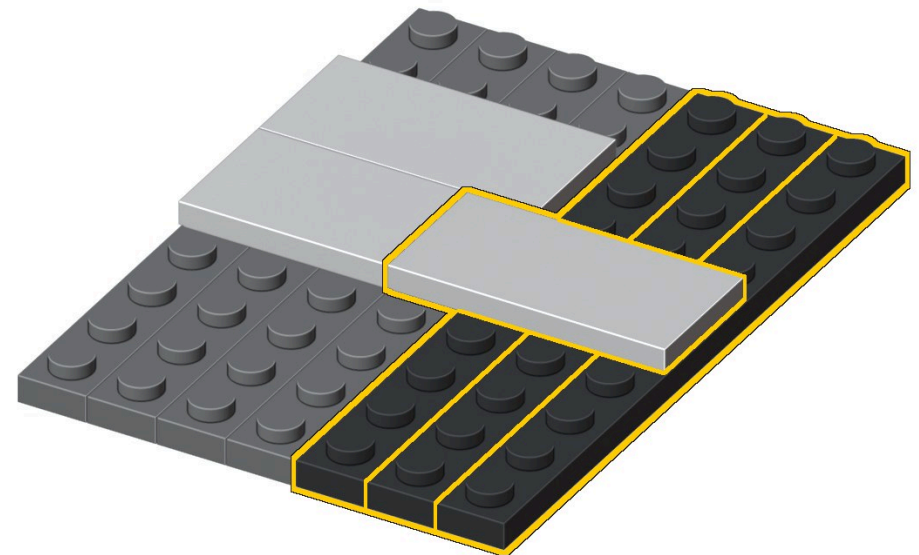
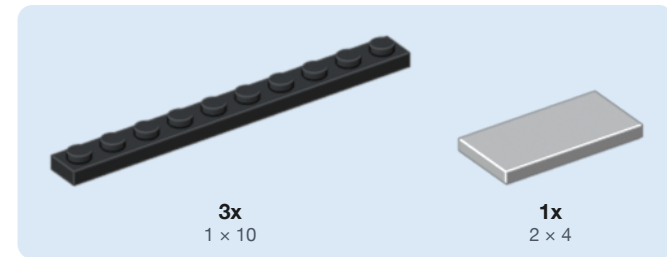


# 107

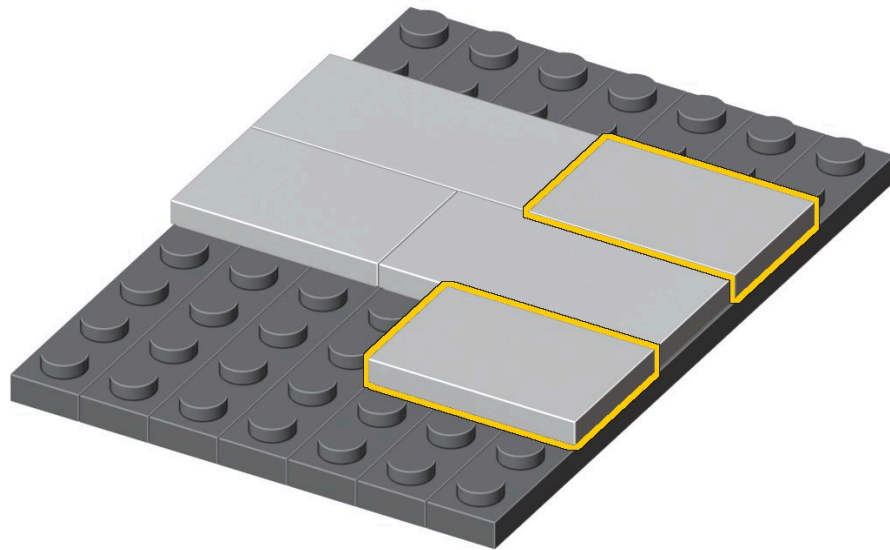
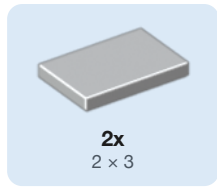


# 108

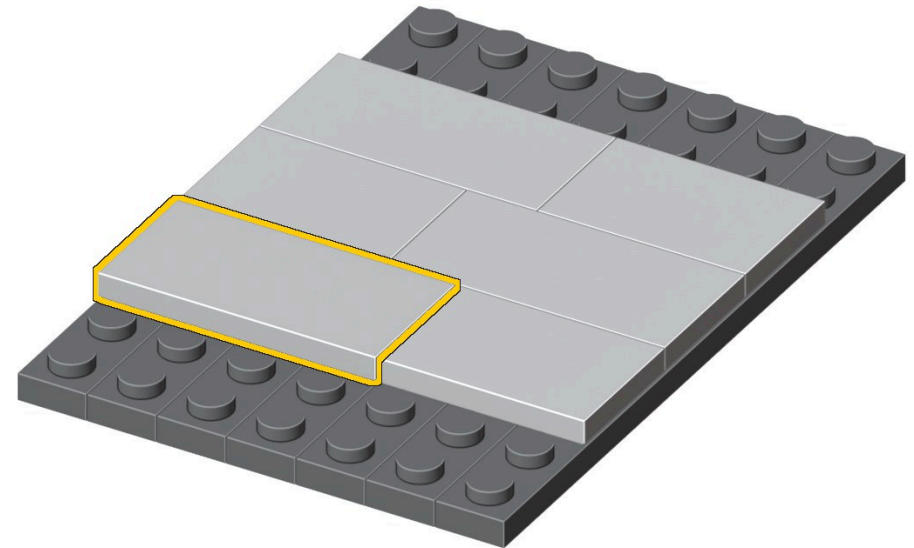
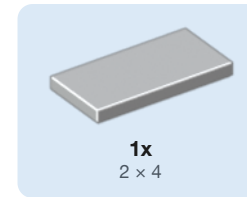
Face label



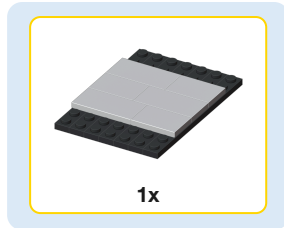
# 109



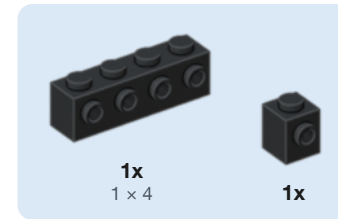
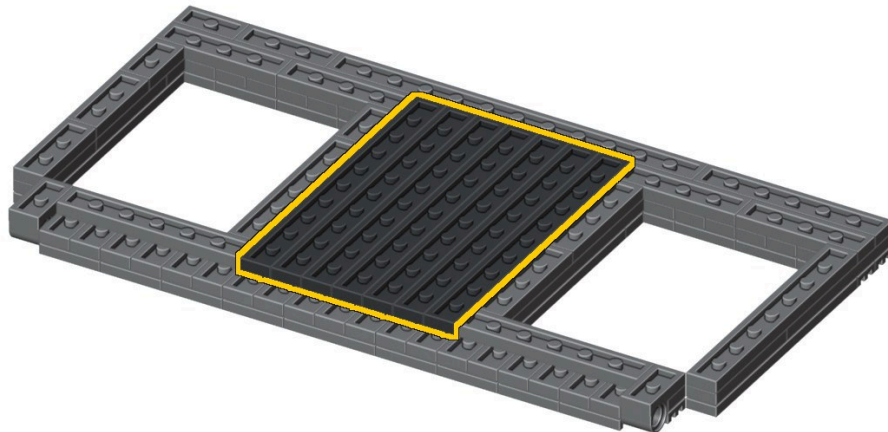
# 110



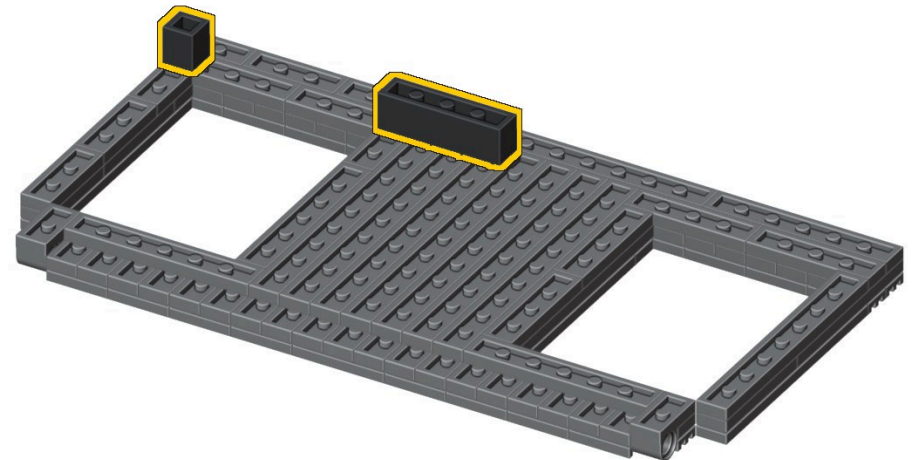
Face label



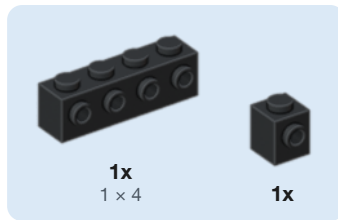
The face label panel: it ties the front and back grip bands together



Bricks with side studs for the spine label



# 113



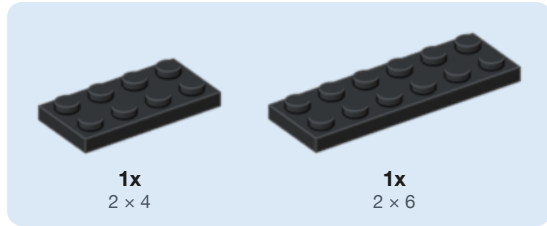
# 114



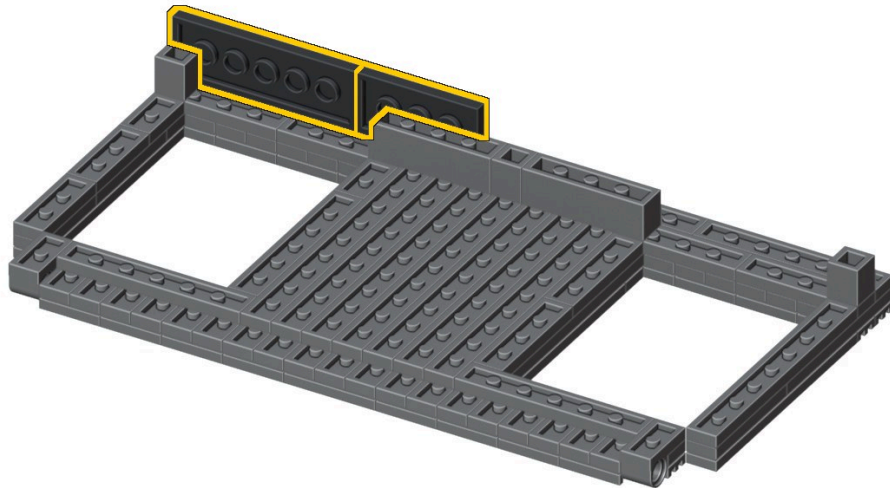
Top shell



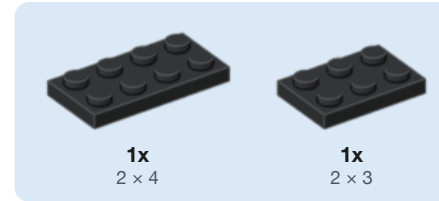
# 115



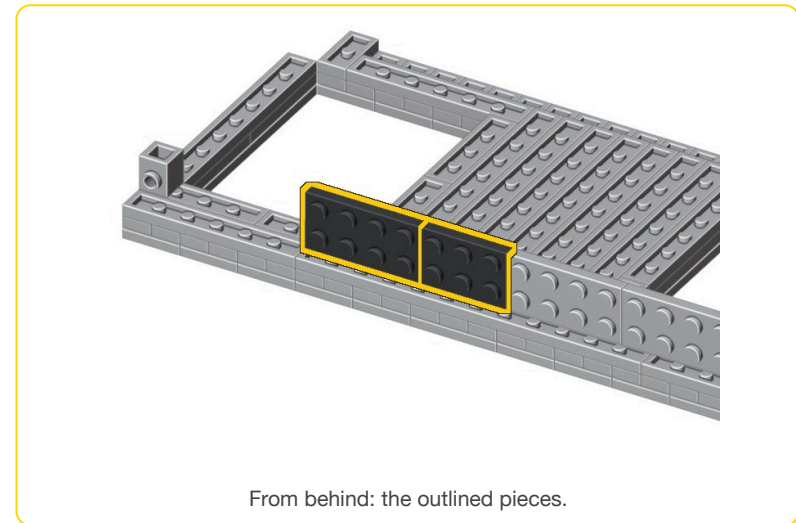
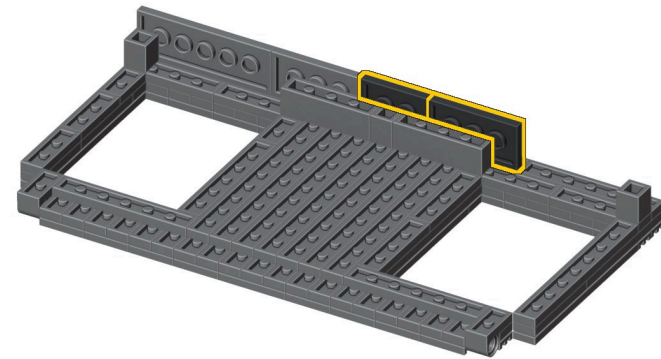
Spine label: plates stood on edge



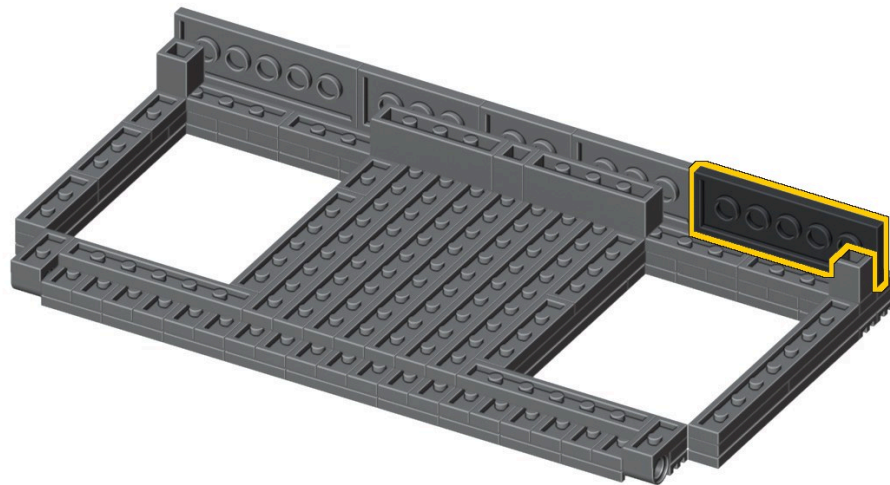
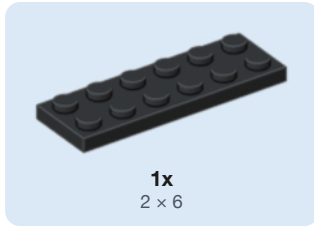
# 116



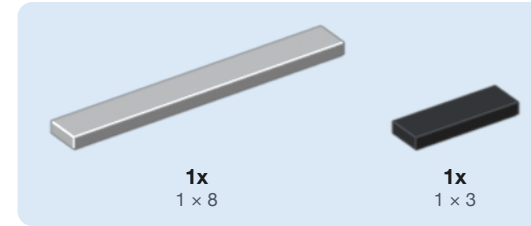
Top shell



# 117



# 118

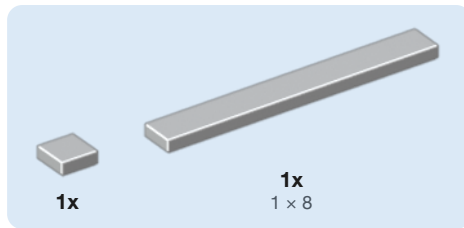


White tiles: the spine label; black tiles at the ends

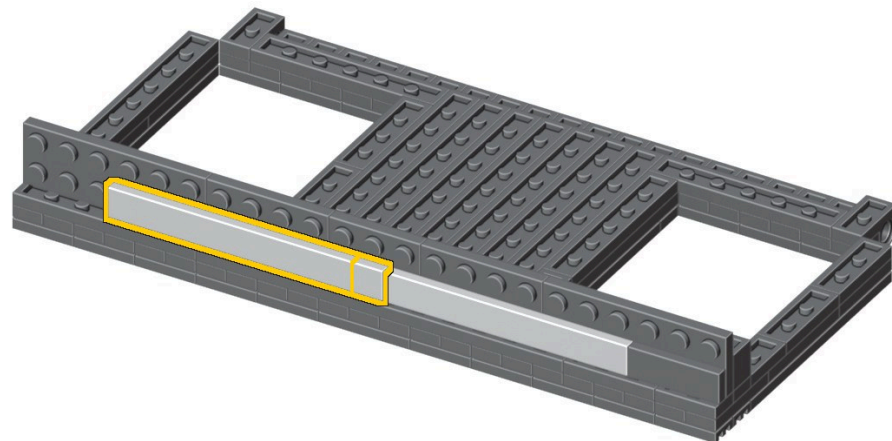
Top shell  
seen from behind



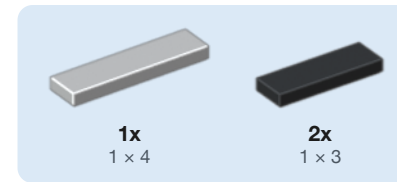
# 119



seen from behind

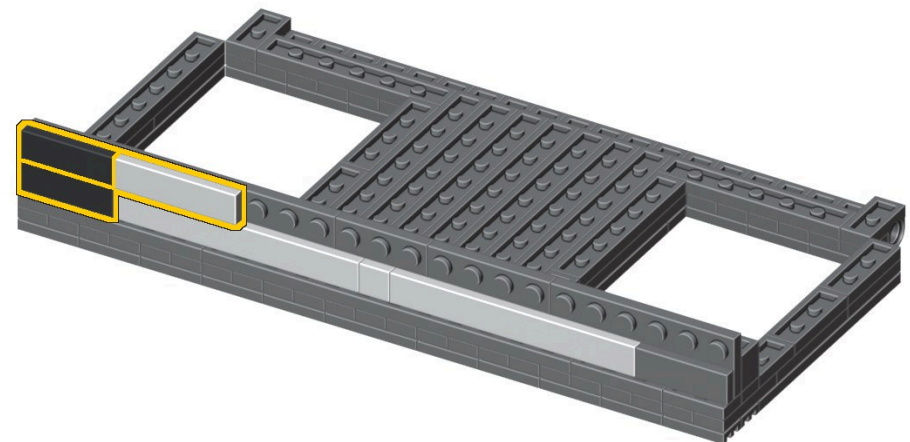


# 120

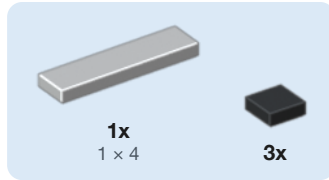


seen from behind

Top shell



# 121

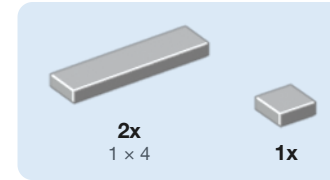


seen from behind

The write-protect tab: a tile you can pull off to protect your recording

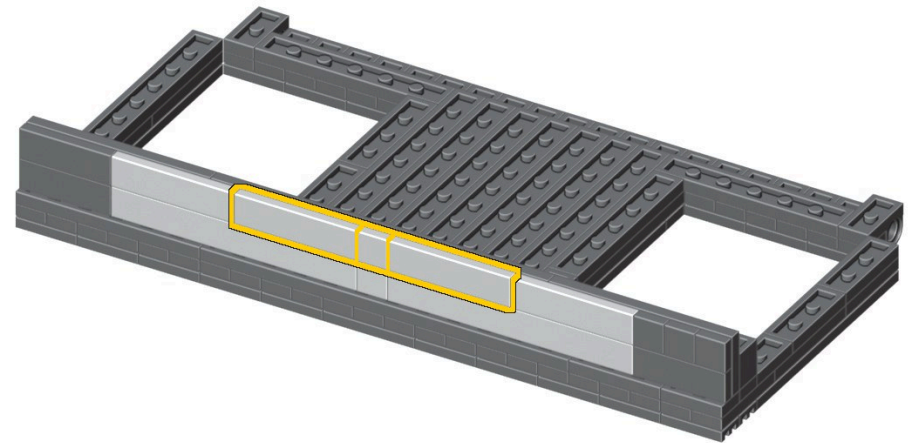


# 122



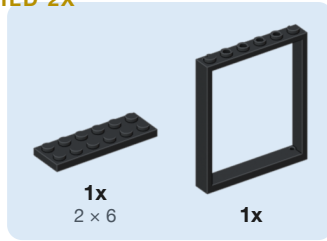
seen from behind

Top shell



WINDOW · BUILD 2X

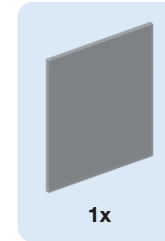
123



A window: a 2 x 6 plate stood on its edge, studs to the front, and the door frame 1 x 6 x 6 laid flat and pushed onto the plate's top row



124

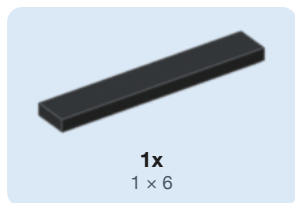


The glass

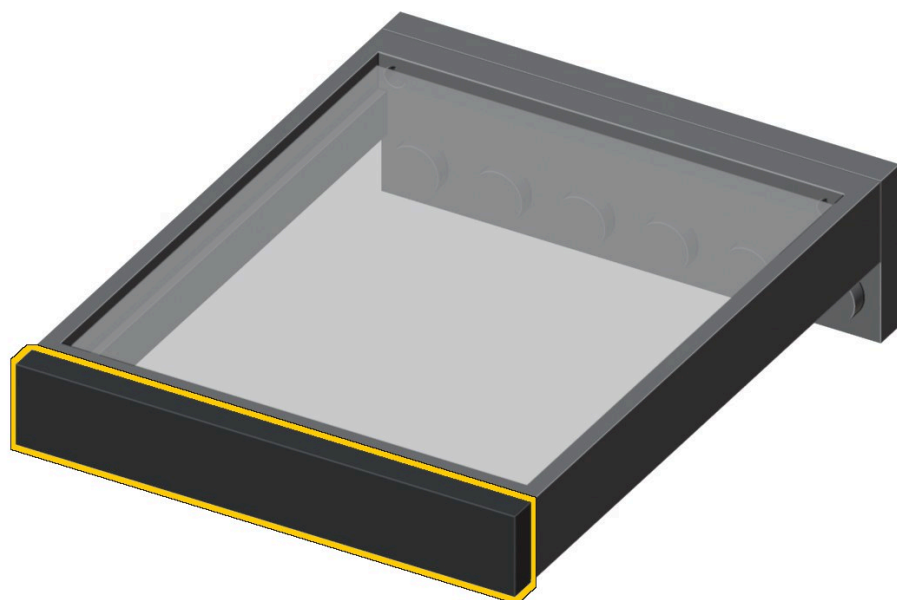


Window

# 125



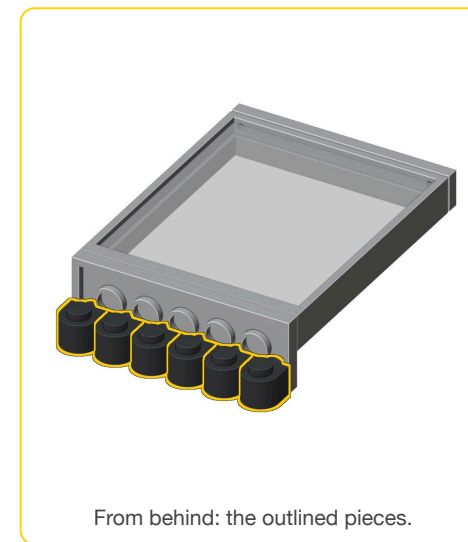
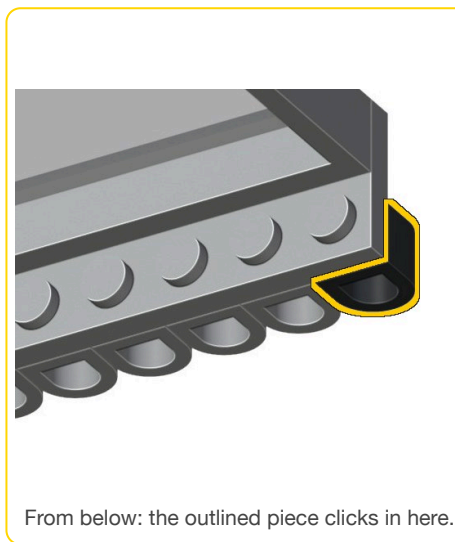
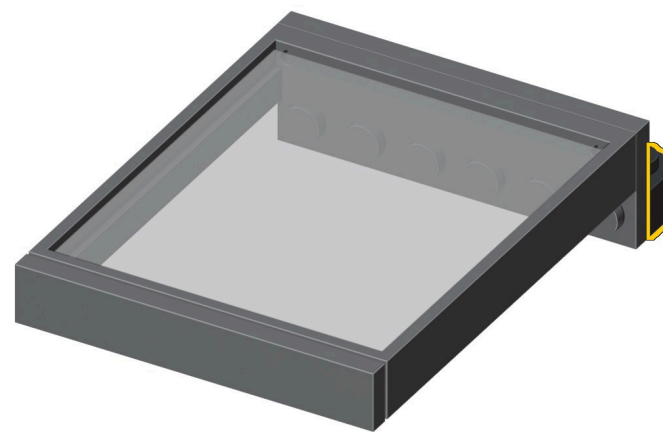
A tile caps the frame's front studs



# 126

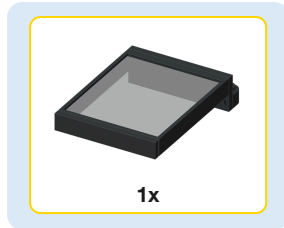


Six hangers on the plate's back row

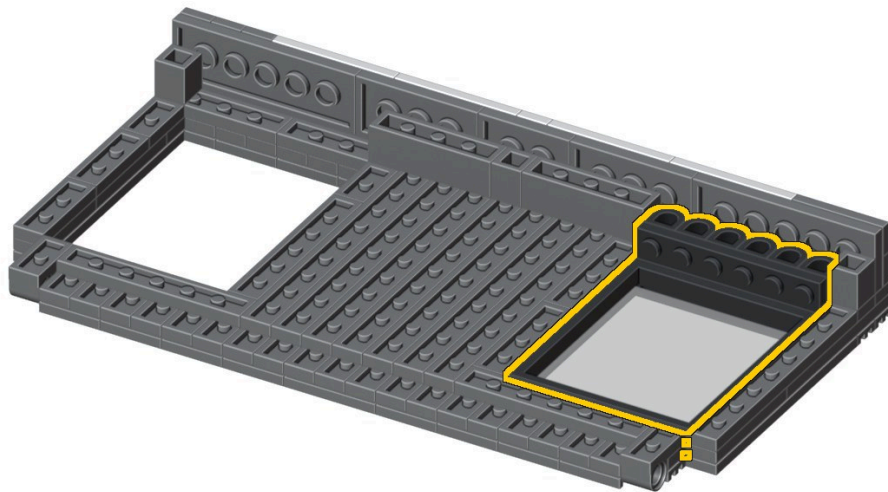


Window

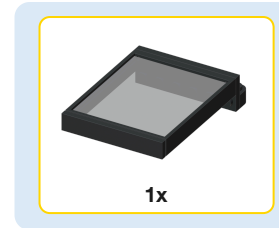
TOP SHELL  
**127**



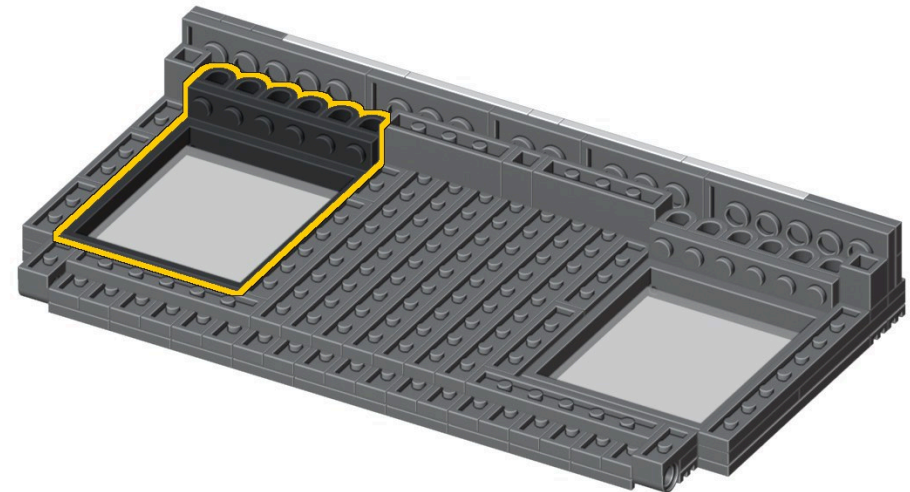
The window, hung on its six hangers



**128**

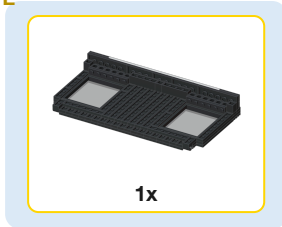


Top shell



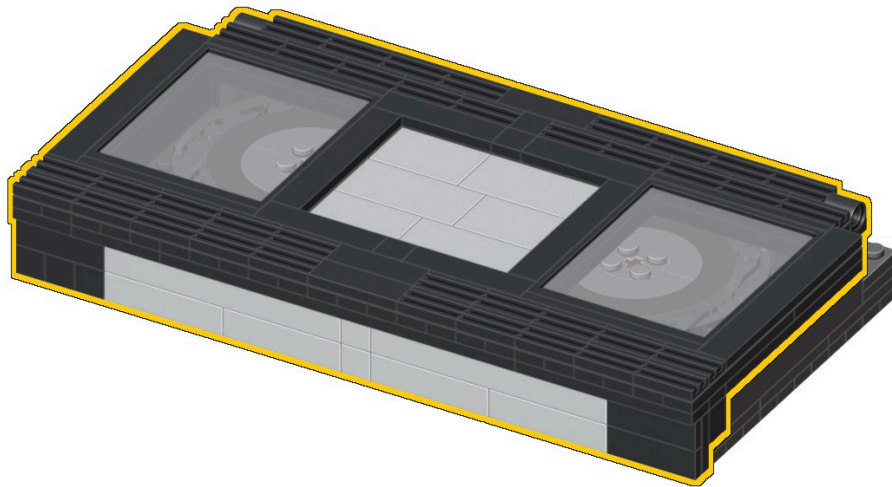
# VHS CASSETTE

## 129



seen from behind

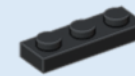
Lower the top shell onto the chassis



# DUST FLAP

## 130

Dust flap



1x  
1 x 3

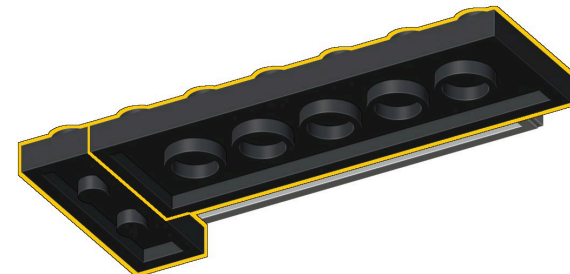
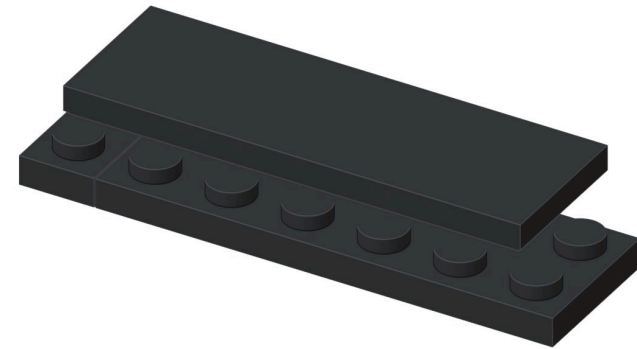


1x  
2 x 6



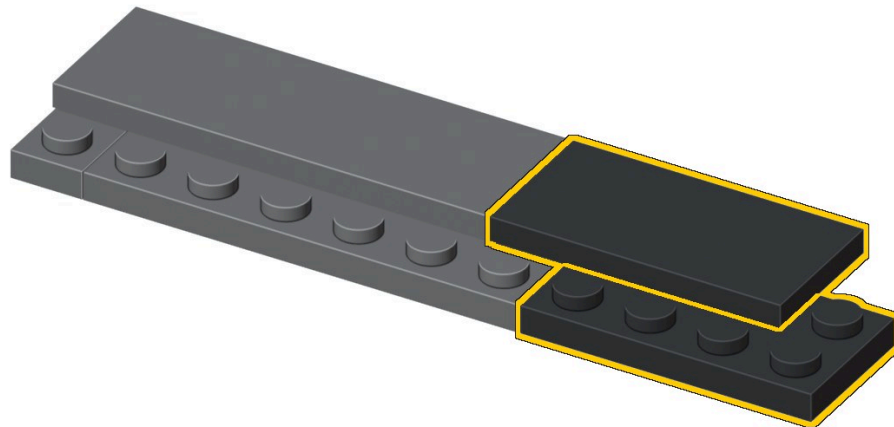
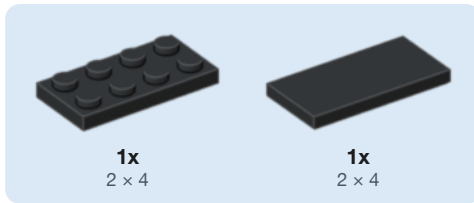
1x  
2 x 6

Dust flap, built flat: 2 x N plates in a row, locked together by 2 x N tiles. The tiles' top halves hang over the plates: they are the flap's top row

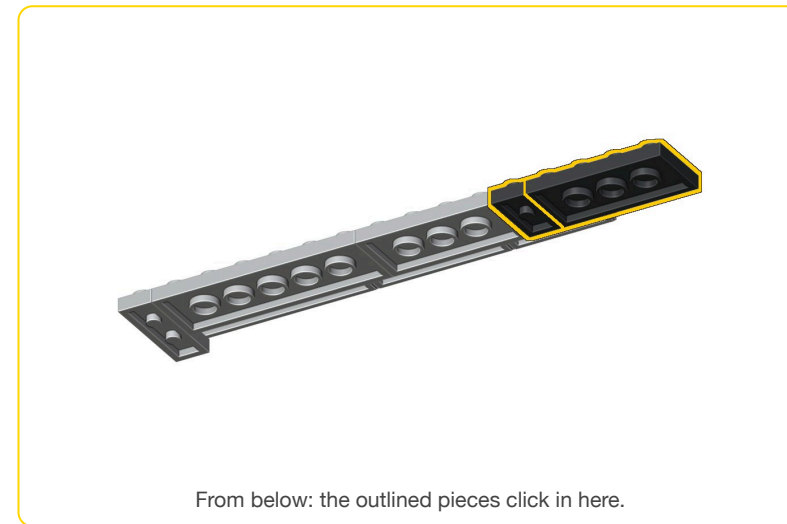
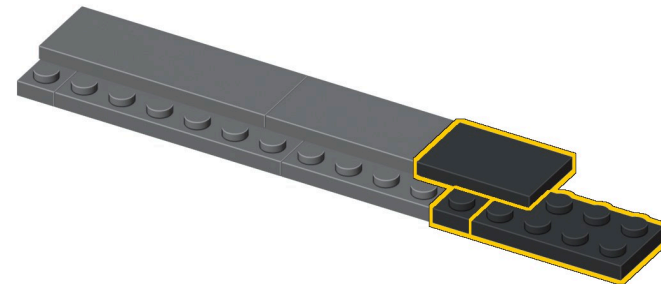
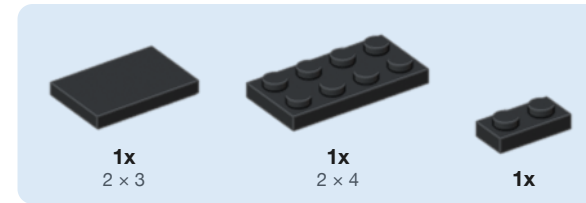


From below: the outlined pieces click in here.

# 131

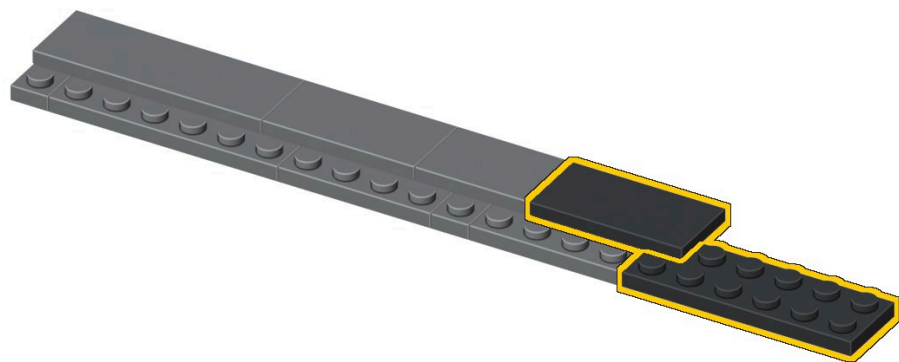
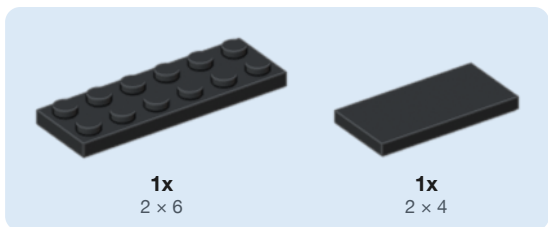


# 132



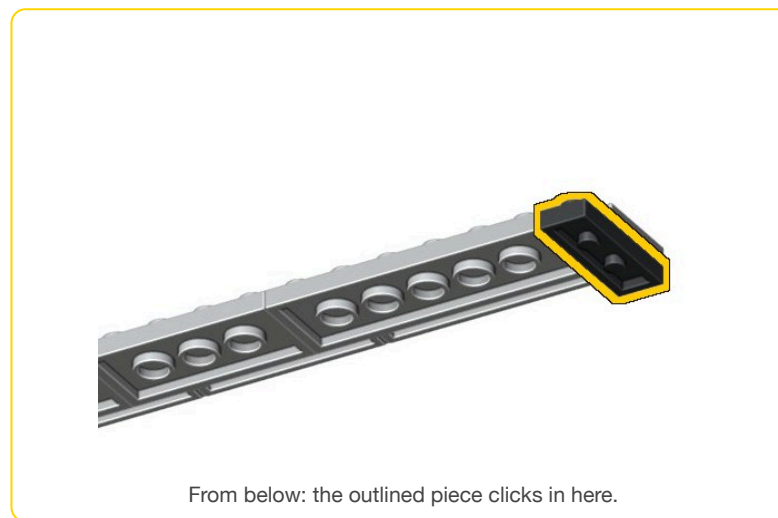
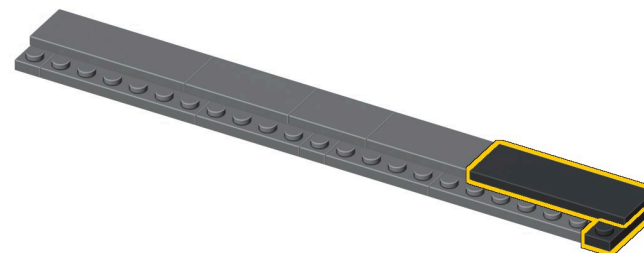
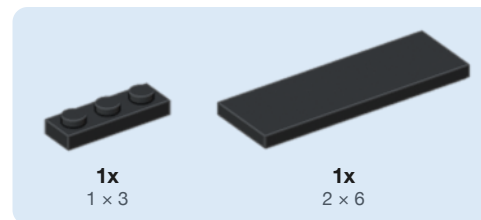
Dust flap

# 133

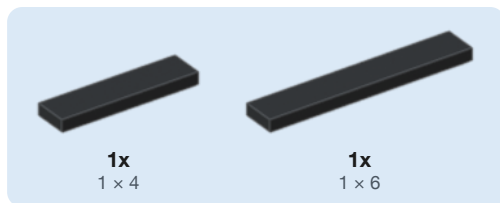


# 134

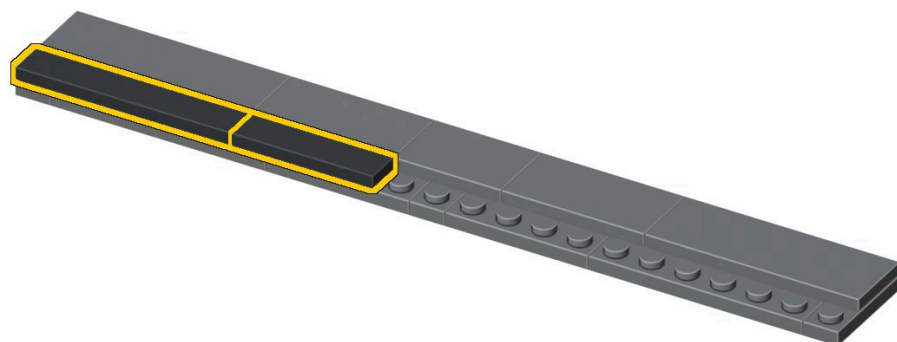
Dust flap



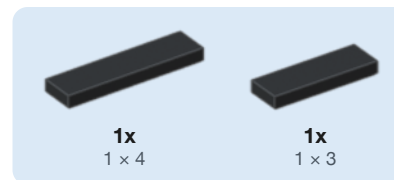
# 135



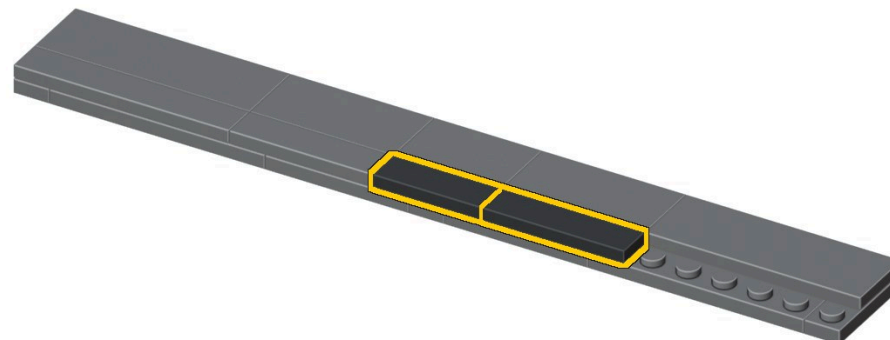
Bottom row of tiles



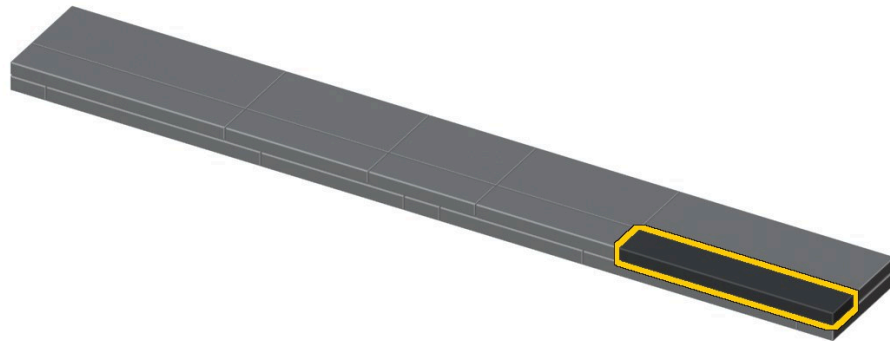
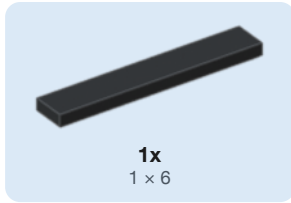
# 136



Dust flap



# 137



# 138



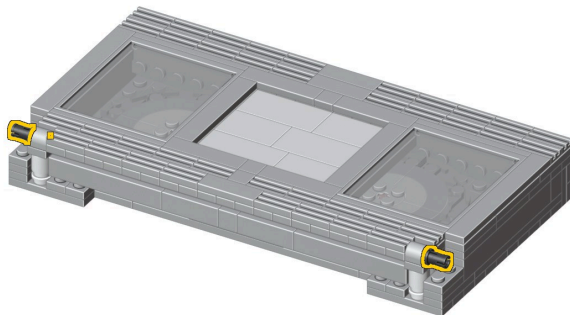
**Dust flap**  
seen from below

Turn it over: Technic bricks at the ends are the flap's cheeks



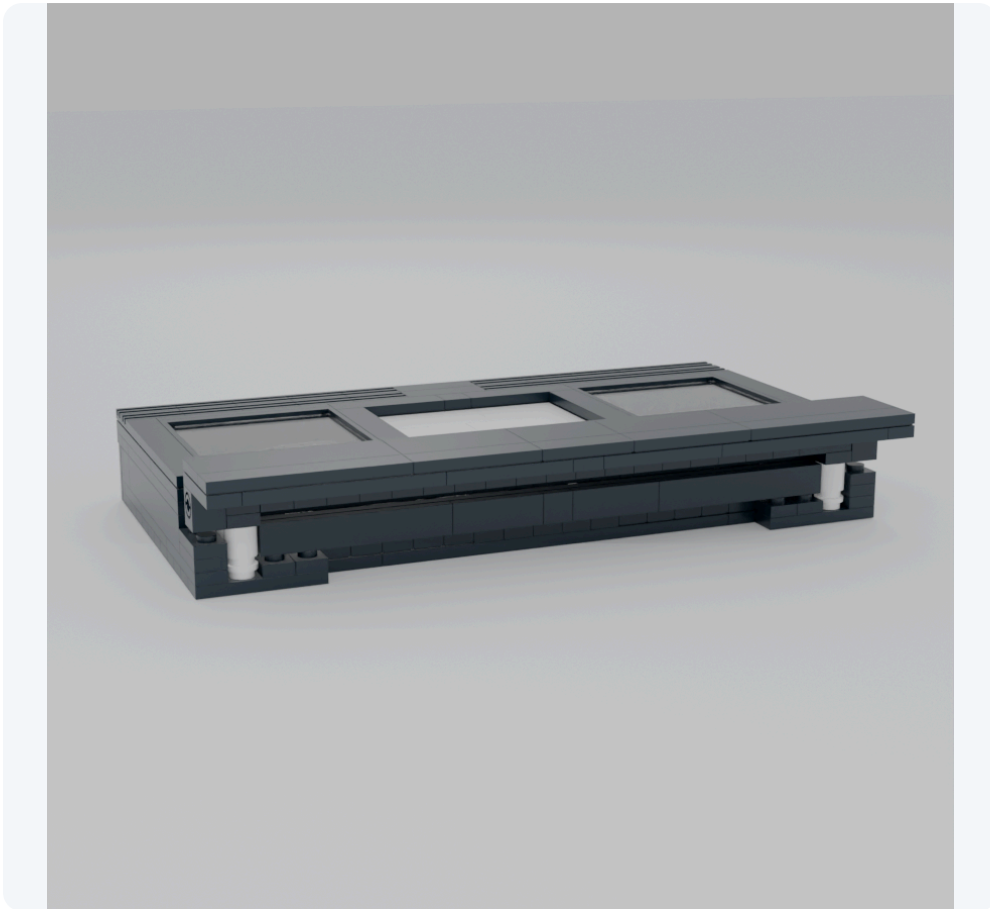


Hold the dust flap in place and push a pin through each cheek



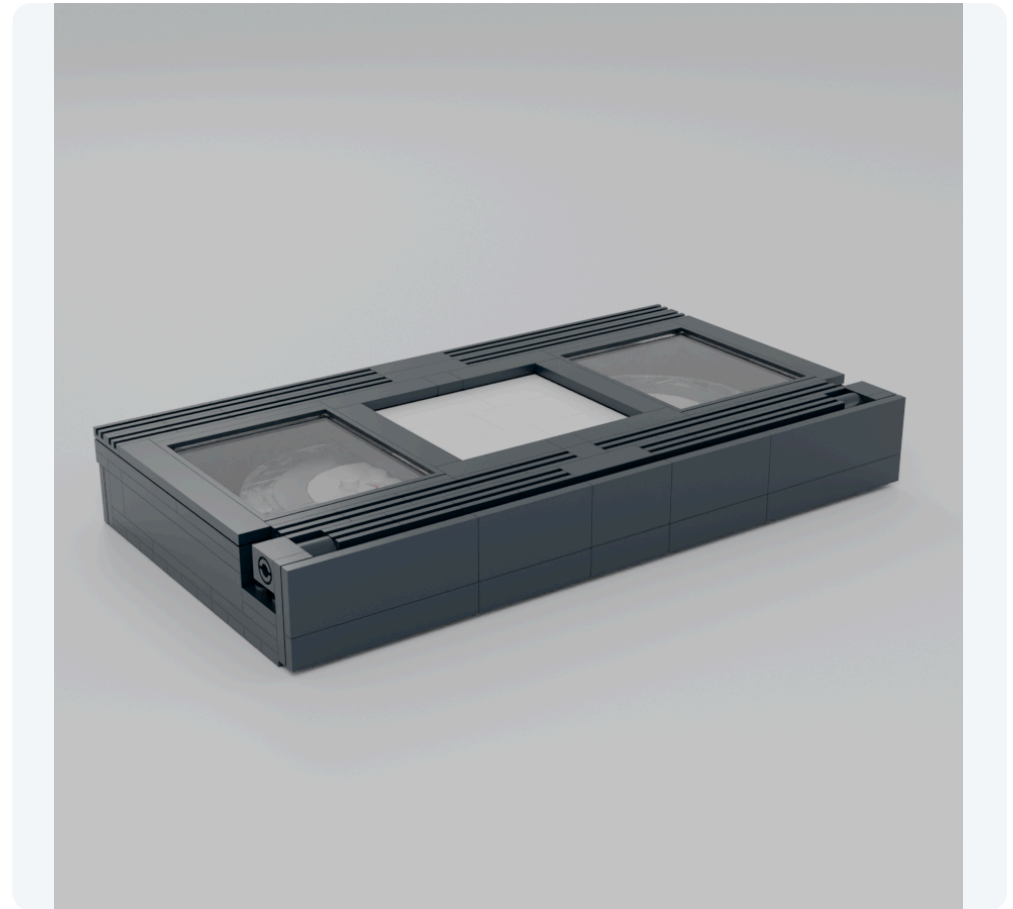
First the outlined pieces: the others go on over them.

# It works!



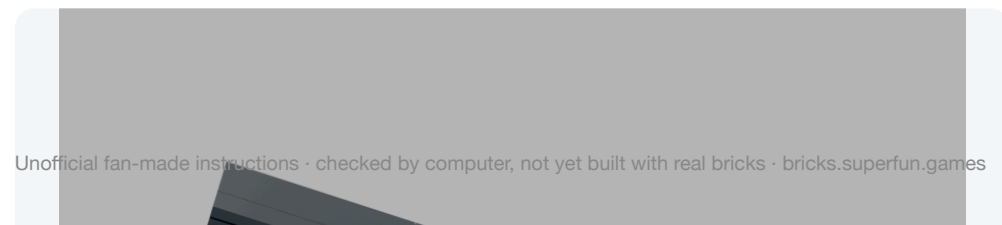
## Dust flap

The whole front edge is the flap. It swings up and out 90 degrees on two Technic pins through the end cheeks, uncovering the tape between two white guide rollers.



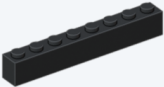
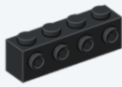
## Turning reels

Each reel turns on a red Technic axle in round holes in the chassis. Watch the supply and take-up packs through the clear windows, or turn a spindle from underneath like a VCR does.



# Parts inventory

384 pieces in 53 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

|                                                                                                                                                |                                                                                                                                                |                                                                                                                                                |                                                                                                                                                 |                                                                                                                                                  |                                                                                                                                                  |                                                                                                                                                       |                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>2x</b><br/>300826<br/>Black<br/>BrickLink 3008</p>     |  <p><b>1x</b><br/>306226<br/>Black<br/>BrickLink 3062b</p>    |  <p><b>1x</b><br/>614326<br/>Black<br/>BrickLink 3941</p>     |  <p><b>6x</b><br/>4558954<br/>Black<br/>BrickLink 87087</p>   |  <p><b>2x</b><br/>4162443<br/>Black<br/>BrickLink 30414</p>   |  <p><b>2x</b><br/>6391342<br/>Black<br/>BrickLink 42205</p>   |  <p><b>12x</b><br/>302426<br/>Black<br/>BrickLink 3024</p>         |  <p><b>9x</b><br/>447726<br/>Black<br/>BrickLink 4477</p>   |
|  <p><b>6x</b><br/>4514845<br/>Black<br/>BrickLink 60479</p>   |  <p><b>48x</b><br/>302326<br/>Black<br/>BrickLink 3023</p>    |  <p><b>48x</b><br/>362326<br/>Black<br/>BrickLink 3623</p>    |  <p><b>24x</b><br/>371026<br/>Black<br/>BrickLink 3710</p>    |  <p><b>12x</b><br/>366626<br/>Black<br/>BrickLink 3666</p>    |  <p><b>8x</b><br/>346026<br/>Black<br/>BrickLink 3460</p>     |  <p><b>1x</b><br/>302126<br/>Black<br/>BrickLink 3021</p>          |  <p><b>4x</b><br/>302026<br/>Black<br/>BrickLink 3020</p>   |
|  <p><b>6x</b><br/>379526<br/>Black<br/>BrickLink 3795</p>     |  <p><b>1x</b><br/>614126<br/>Black<br/>BrickLink 6141</p>     |  <p><b>1x</b><br/>403226<br/>Black<br/>BrickLink 4032a</p>    |  <p><b>4x</b><br/>6393938<br/>Black<br/>BrickLink 74611</p>   |  <p><b>12x</b><br/>6424674<br/>Black<br/>BrickLink 3386</p>   |  <p><b>2x</b><br/>6114987<br/>Black<br/>BrickLink 11458</p>   |  <p><b>2x</b><br/>654126<br/>Black<br/>BrickLink 6541</p>          |  <p><b>2x</b><br/>4121715<br/>Black<br/>BrickLink 2780</p>  |
|  <p><b>1x</b><br/>6280168<br/>Black<br/>BrickLink 32124</p>  |  <p><b>3x</b><br/>370926<br/>Black<br/>BrickLink 3709</p>    |  <p><b>7x</b><br/>307026<br/>Black<br/>BrickLink 3070b</p>   |  <p><b>4x</b><br/>306926<br/>Black<br/>BrickLink 3069</p>    |  <p><b>21x</b><br/>4558170<br/>Black<br/>BrickLink 63864</p> |  <p><b>14x</b><br/>243126<br/>Black<br/>BrickLink 2431</p>   |  <p><b>14x</b><br/>663626<br/>Black<br/>BrickLink 6636</p>        |  <p><b>2x</b><br/>416226<br/>Black<br/>BrickLink 4162</p>  |
|  <p><b>1x</b><br/>6162892<br/>Black<br/>BrickLink 26603</p> |  <p><b>2x</b><br/>4560182<br/>Black<br/>BrickLink 87079</p> |  <p><b>2x</b><br/>6318582<br/>Black<br/>BrickLink 69729</p> |  <p><b>8x</b><br/>6434275<br/>Black<br/>BrickLink 79393</p> |  <p><b>38x</b><br/>241226<br/>Black<br/>BrickLink 2412b</p> |  <p><b>2x</b><br/>6143850<br/>Black<br/>BrickLink 15535</p> |  <p><b>5x</b><br/>4633691<br/>Flat Silver<br/>BrickLink 6141</p> |  <p><b>2x</b><br/>4142865<br/>Red<br/>BrickLink 32062</p> |

# Parts inventory (continued)

384 pieces in 53 part/colour lines. Bold: the LEGO element ID (what LEGO Pick a Brick sells); below: the colour and the BrickLink part number.

|                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>2x</b><br><b>6511024</b><br> Trans-Clear<br>BrickLink 42509 | <br><b>8x</b><br><b>6544432</b><br> Trans-Clear<br>BrickLink 27507 | <br><b>3x</b><br><b>306201</b><br> White<br>BrickLink 3062b  | <br><b>1x</b><br><b>302301</b><br> White<br>BrickLink 3023   | <br><b>1x</b><br><b>362301</b><br> White<br>BrickLink 3623   | <br><b>3x</b><br><b>614101</b><br> White<br>BrickLink 6141 | <br><b>2x</b><br><b>403201</b><br> White<br>BrickLink 4032a | <br><b>2x</b><br><b>307001</b><br> White<br>BrickLink 3070b |
| <br><b>4x</b><br><b>243101</b><br> White<br>BrickLink 2431         | <br><b>2x</b><br><b>416201</b><br> White<br>BrickLink 4162         | <br><b>8x</b><br><b>6172674</b><br> White<br>BrickLink 27925 | <br><b>3x</b><br><b>6156667</b><br> White<br>BrickLink 26603 | <br><b>3x</b><br><b>4560178</b><br> White<br>BrickLink 87079 |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                   |

# About this model

Built like a real cassette, from the inside out. First the chassis: the bottom with its drive holes and screw heads, the lamp tube, the reel-lock release, the guide rollers and the tape stretched across the mouth. Two reels drop onto their red spindles: the supply reel and the take-up reel. The end walls go on, then the top shell, built upside down with its grip bands, recessed label, spine label and two big windows, is lowered over everything. Last, the full-length dust flap hangs on two Technic pins.

## Checked on a computer

|      |               |                                                                            |
|------|---------------|----------------------------------------------------------------------------|
| PASS | real elements | 53 part/colour combinations: 0 not real, 0 rare                            |
| PASS | connections   | 996 connections (965 stud, 21 pin, 6 axle, 4 hinge); 1 separate piece(s)   |
| PASS | collisions    | 0 overlapping pair(s) among 384 parts                                      |
| PASS | buildability  | 262 insertions tried across 14 submodel(s); 0 problem(s)                   |
| PASS | stability     | about 313 g; centre of mass inside the base (98% margin); tips at 74.9 deg |
| PASS | mechanism     | 24 poses swept for 3 moving group(s); 0 gear pair(s); 0 problem(s)         |
| PASS | electrics     | no electrics in this model                                                 |
| PASS | technique     | 0 note(s)                                                                  |

## Not yet built with real bricks

These checks catch most mistakes a designer can make, but not all: clutch power, part tolerances, how stiff a long axle is and how the model survives handling can only be judged with real bricks. If you build it, we would love to hear how it went.

## Files

3D model (LDraw, opens in BrickLink Studio, LeoCAD and LDCad), parts lists for BrickLink and LEGO Pick a Brick, and an interactive 3D viewer:

**[bricks.superfun.games/m/vhs\\_tape](https://bricks.superfun.games/m/vhs_tape)**

## Trademarks

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